

Review

The influence of RE elements on the performance of laser welded joints of magnesium alloys: A review

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Abstract

Among existing lightweight metals, magnesium (Mg) alloys have garnered significant attention due to their exceptional specific strength. However, their laser welding applications face challenges from porosity, cracking, and grain coarsening defects. Rare earth (RE) elements, leveraging their unique strengthening effects, offer a promising solution for refining weld microstructures and suppressing welding defects. Nevertheless, a systematic review of RE-enhanced mechanisms and defect suppression strategies remains lacking. This paper systematically reviews recent research advancements in Mg alloy laser welding, with a focused elucidation of the governing effects of welding parameters on weld performance, and the core mechanistic roles of RE elements in the welded joint. Furthermore, we discuss key challenges and future directions in process optimization, service performance enhancement, and industrial scalability of RE-modified Mg alloy welding. The findings aim to provide theoretical foundations for designing high-performance welded Mg-RE structures and advance lightweight manufacturing technologies in aerospace, electric vehicles, and other cutting-edge industries.

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1. Introduction

As one of the lightest structural engineering materials, magnesium (Mg) alloys exhibit remarkable application potential in aerospace, automotive manufacturing, electronic devices, and biomedical fields [1–3]. This advantage stems from their exceptional specific strength (~ 130 kN·m/kg), low density (~ 1.738 g/cm³), and superior castability. However, intrinsic limitations including inadequate room-temperature strength, compromised high-temperature creep resistance, and limited weldability substantially constrain their implementa-

tion in critical load-bearing applications [4–6]. Welding, being a pivotal fabrication technology for Mg components, directly governs structural integrity through joint performance. The combination of Mg alloys' relatively low melting point (~ 650 °C), pronounced oxidation propensity, and rapid solidification behavior frequently induces weld defects such as porosity, crack formation, and grain coarsening during joining operations [7–9].

The morphology and distribution of strengthening phases critically govern the performance enhancement of welded joints. Shen et al. [10] investigated welding speed effects on AZ31 alloy processing, revealing that increasing welding speed refined fusion zone (FZ) grains while enhancing β -Mg₁₇Al₁₂ particle volume fraction, consequently improving ultra-tensile strength (UTS), yield strength (YS), and elongation (EL). Through laser-TIG hybrid welding of AZ31B and AZ80 alloys respectively, Song et al. [11] achieved welded

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joints with homogenized β -Mg₁₇Al₁₂ phase distributions and refined microstructures, demonstrating joint effectiveness factors exceeding 100 %. Contrastingly, S. Srinivasan et al. [12] observed continuous β -Mg₁₇Al₁₂ network formation under 504 J/mm heat input conditions, inducing weakened intergranular bonding strength and significant solidification cracking. To suppress weld cracking, Zai et al. [13] incorporated MgO into TIG filler wires. The favorable lattice compatibility between MgO and α -Mg matrix promoted fully equiaxed grain formation in FZs, achieving 65.8 % grain size reduction that substantially mitigated joint hot cracking susceptibility. During laser remelting experimental research on die-cast AZ91D alloy, Shen et al. [14] identified hydrogen porosity as the predominant void type in remelted regions. Mikucki et al. [15] demonstrated that hydrogen precipitation behavior triggered by solubility variations during rapid temperature decline constitutes the primary hydrogen porosity mechanism in AZ91 alloys, with porosity levels exhibiting linear proportionality to hydrogen content.

The incorporation of RE elements (e.g., Gd, Y, Ce, Nd) offers novel pathways for enhancing weldability in Mg alloys. Established research indicates RE elements can substantially improve both ambient and elevated-temperature mechanical properties through three primary mechanisms: formation of high-melting-point intermetallics (e.g., Al₄Ce, Mg₂₄Y₅), grain refinement, and controlled precipitation distribution [16,17]. However, RE interactions during welding exhibit greater complexity. In automotive-grade ME21 alloy welding research by Li et al. [18], cerium (Ce) induced second phases strengthening via Al₃Ce intermetallic formation, achieving 81.8 % joint efficiency with UTS reaching 185.2 MPa. Chen et al. [19,20] revealed that high-melting-point intermetallics (e.g., Al₁₇Ce₂Mn₇, Al₁₁Ce₃) formed via Ce-Al interactions effectively pin grain boundaries (GBs), reducing HAZ GBs migration rates and suppressing heat-affected zone (HAZ) grain coarsening. Yan et al. [21] demonstrated in Mg-3Nd-0.2Zn-0.4Zr alloy welding that the heterogeneous nucleation effects of neodymium reduced weld zone grain size by 76 %, concurrently increasing hardness from 53 HV_{0.2} to 58 HV_{0.2}. Notably, the welded joints retained 71 % of room-temperature UTS at 200 °C, attributable to thermally stable Mg₁₂Nd precipitates.

Laser welding, as a high-energy-density joining technology, demonstrates superior compatibility with Mg alloys' low melting characteristics through its inherent advantages of low heat input, high depth-to-width ratio, and narrow HAZ [22,23]. M. Vyskoč et al. [24] demonstrated that reducing laser power from 2 kW to 1.2 kW during AZ31 alloy welding accelerated cooling rates, refining FZ equiaxed grains from 29.59 μ m to 20.51 μ m. This process enhancement resulted in 28.4 % UTS improvement, 38 % YS increase, 6.4 % hardness elevation, and the joint efficiency was as high as 90 %. However, laser deep penetration welding introduces critical challenges including keyhole instability, elemental vaporization, and non-equilibrium solidification, which exacerbate microstructural heterogeneity and defect susceptibility. M. Courtois et al. [25] established through three-dimensional (3D)

modeling that cavity entrapment from keyhole collapse primarily governs porosity formation, with porosity content following exponential growth relative to heat input. Hu et al. [26] developed an adjustable ring-mode (ARM) laser welding model revealing 75 % porosity reduction when melting pool/keyhole oscillation amplitude doubled (100 % increase) and the oscillation period extended by 64 %. Zhou et al. [27] conducted numerical simulations of thermo-physical fields and microstructural evolution during ZW61 alloy welding. Their findings revealed that increased welding speed significantly elevates temperature gradients within the molten pool, while asynchronous cooling across material contraction zones intensifies residual stress accumulation. In laser welding research on AA7075 aluminum (Al) alloy by M. Holzer et al. [28], energy-dispersive spectroscopy (EDS) measurements detected elemental concentration variations at weld seams. The study demonstrated that high energy density induces compositional fluctuations in welded joints, which directly influence hot cracking susceptibility.

Current research on laser welding of RE-free Mg alloys primarily emphasizes process parameter optimization (including laser power, welding speed, and focal position) and their influences on weld morphology and mechanical performance. In contrast, welding investigations of Mg-RE alloys predominantly concentrate on RE elements' modulation mechanisms governing non-equilibrium solidification microstructures, second phases evolution, and defect formation, particularly grain refinement, GB pinning, and second phases control [29–33]. Although existing studies confirm that RE elements enhance joint strength via nanoscale precipitate formation, critical knowledge gaps persist regarding the evaporation characteristics of RE elements under high-energy-density laser welding conditions, solute segregation patterns within molten pools, and the behavior among dissolution-precipitation dynamics of RE compounds during thermal cycling, microstructural evolution, and defect mitigation mechanisms. Although existing literature has reviewed the advancements in laser welding of magnesium alloys, most studies compartmentalize the introduction of RE elements and the laser welding process as distinct domains of inquiry [34–39]. They often fail to delve into the unique non-equilibrium solidification kinetics and defect formation mechanisms induced by high-energy-density lasers. Furthermore, research seldom systematically elucidates the role of RE elements in these critical processes. Consequently, a systematic analysis of the interaction between RE elements and the laser welding of magnesium alloys remains notably lacking.

This paper systematically examines the pivotal role of RE elements in laser welding of Mg alloys, focusing on their mechanistic influences on weld joint microstructures, mechanical properties, and defect formation. The analysis initiates with laser welding characteristics and their compatibility with Mg alloys, followed by detailed elucidation of RE elements strengthening mechanisms in Mg-RE systems encompassing grain refinement, GBs pinning, and second phases evolution. Subsequent sections address defect generation mechanisms and RE elements regulatory effects, concluding with perspec-

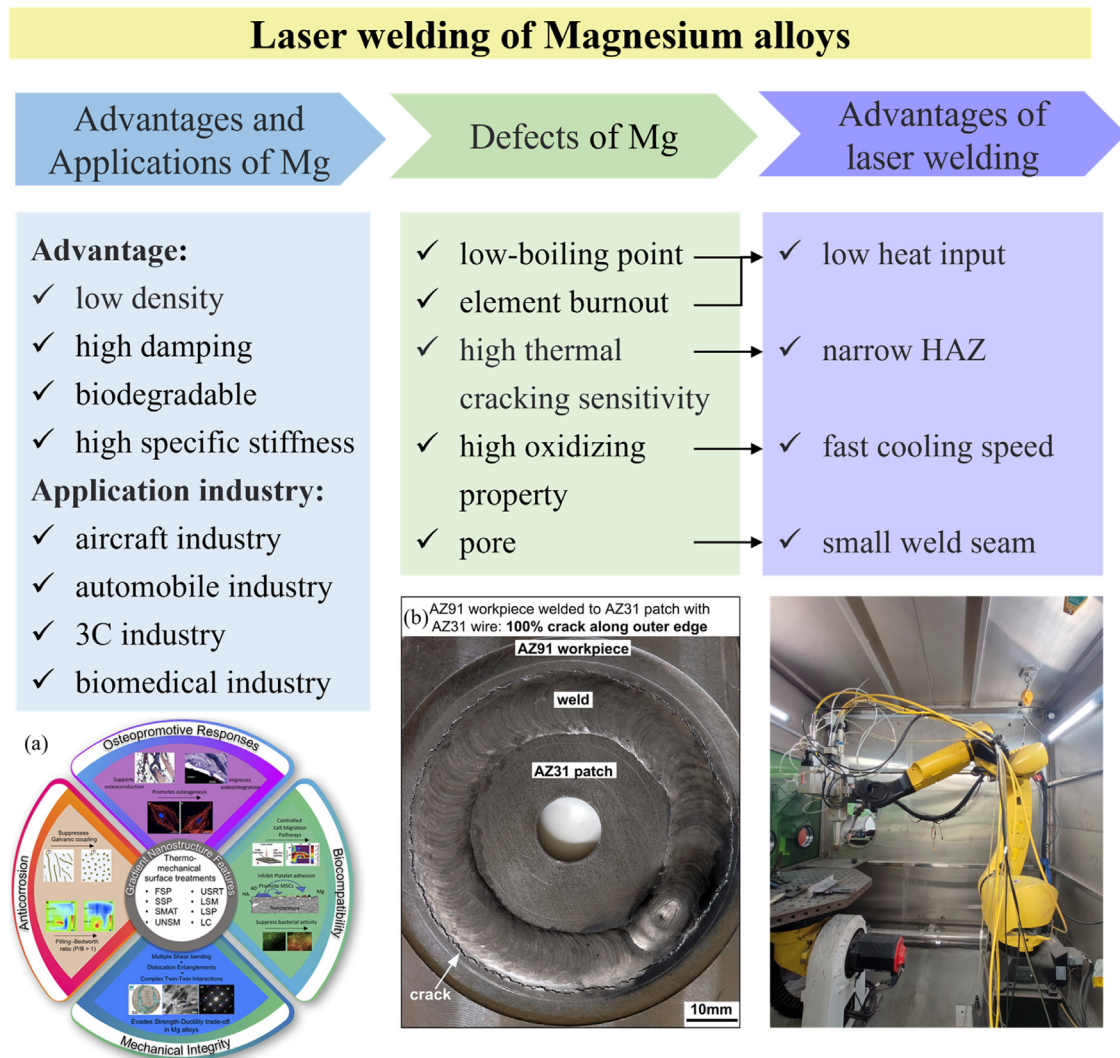


Fig. 1. Compatibility analysis of Mg alloys and laser welding technology [41,42].

tives on challenges and future directions in RE-containing Mg alloy laser welding. The findings provide theoretical foundations for engineering applications of high-performance welded Mg structures in lightweight industries.

2. Laser welding characteristics and compatibility with Mg alloys

2.1. Classification of laser welding modes

Fig. 1 presents a synergistic analysis of Mg alloy compatibility with laser welding technology. Conventional welding of Mg alloys exhibits pronounced limitations including high oxidation susceptibility, hot cracking sensitivity, and porosity formation. Laser welding, leveraging its ultrahigh power density, rapid thermal cycling, and precise heat input control, effectively mitigates these challenges [40]. This technique substantially constricts the HAZ, suppresses surface oxidation and elemental vaporization, and enables stringent molten pool dy-

namic behavior governance. Consequently, it reduces porosity and crack propensity while yielding joints with enhanced densification and superior mechanical integrity.

During the laser welding process, welding modes can be classified into thermal conductivity welding mode and deep penetration welding mode based on the heat transfer mode and the presence or absence of keyholes. Fig. 2(a-b) schematically illustrate these two distinct welding modalities. In the thermal conductivity welding mode, the fusion of the weld seam is mainly achieved through the heat conduction of the molten pool. A hemispherical, wide, and shallow weld bead is formed on the cross-section of the weld seam, with a depth-to-width ratio of approximately one, similar to the traditional fusion welding process [43]. This regime exhibits elevated surface-area-to-volume ratios ($A/V \propto 1/h$, where h denotes penetration depth) that markedly intensify elemental vaporization. In addition, the stabilized molten pool surface minimizes oxide formation and gas entrapment, thereby exhibiting enhanced resistance to imperfection formation. These characteristics render thermal conductivity welding mode particularly

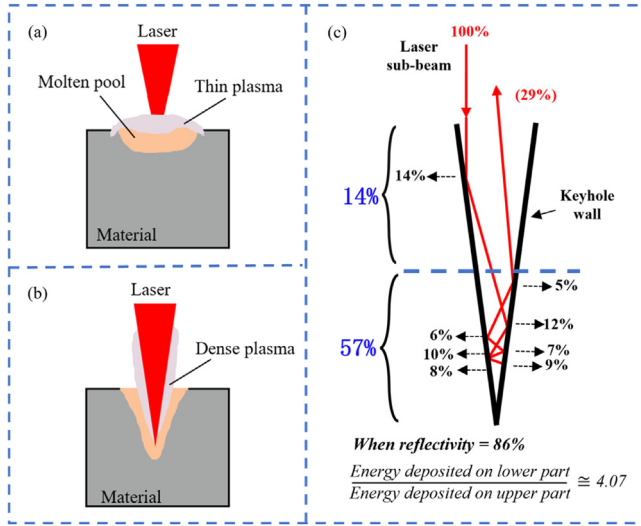


Fig. 2. (a) Thermal conductivity welding mode; (b) deep penetration welding mode; (c) schematic diagram of absorbed energy distribution on the inner wall of the welding keyhole under a material laser reflectivity of 86 % [44].

suitable for joining thin-gauge materials or precision components requiring minimal defect tolerance.

In contrast, the deep penetration welding model operates under high energy density laser irradiation, generating narrow, deeply penetrating weld pools characterized by distinctive keyhole morphologies with depth-to-width ratio exceeding 1.2 [43,45, 46]. Fig. 2(c) schematically illustrates the energy distribution profile within such keyhole structures during deep penetration welding [44]. The enhanced depth-to-width ratio fundamentally arises from improved energy coupling efficiency through multiple laser beam reflections along the keyhole's interior surfaces. The interaction between the laser beam and keyhole wall fundamentally governs iterative Fresnel absorption cycles, driving a substantial enhancement in energy coupling efficiency. This constitutes the pivotal mechanism enabling the elevation of Mg alloys' laser absorptivity from 8 %~20 % to industrial-grade requirements [44,47–50].

The deep penetration welding mode demonstrates dual advantages: achieving rapid penetration through high energy density while maintaining superior welding velocities and processing efficiency. However, this technique is inherently constrained by pronounced keyhole instability, where elevated heat input exacerbates molten pool thermal gradients and induce keyhole fluctuation phenomena, resulting in a more complex welding mechanism.

In addition, the interdependent relationship between Mg alloys' high reflectivity characteristics, laser operational parameters, and keyhole dynamic stability remains insufficiently quantified. It is necessary to further explore key issues to establish fundamental correlations among joint performance metrics, heat input management, alloy composition optimization, and interfacial reaction control. Resolving these scientific challenges is essential for advancing the theoretical framework of Mg alloy laser keyhole welding and accelerating its industrial implementation.

2.2. The influence of welding parameters on Mg alloy welding

The penetration depth and width of the weld seam, joint defects and mechanical properties, etc. are fundamentally governed by welding parameters [30]. Primary laser welding parameters encompass laser power P , welding speed v , defocus amount Δf , shielding gas flow rate, etc.. Among them, laser power and welding speed jointly determine the heat input per unit area. To systematically elucidate the mechanistic influence of these parameters on joint properties, Eq. (1) is adopted to normalize the parameter, quantifying heat input (E) as the principal control variable, as mathematically expressed in Eq. (1):

$$E = (P/v) \cdot \eta \quad (1)$$

Where E is the heat input of laser welding (J/mm), P is the laser power (kW), v is the welding speed (m/min), and η is the welding efficiency. $\eta = 100 \%$ is set to eliminate the influence of equipment differences.

The rapid solidification characteristics inherent to laser welding induce substantial deviation from equilibrium conditions during phase transformation. This non-equilibrium solidification regime promotes pronounced intergranular/intragranular segregation, metastable phase formation, and localized stress intensification, consequently increasing crack susceptibility. The rapid cooling of the molten pool duration t exhibits direct thermodynamic correlation with E , as mathematically formulated in Eq. (2):

$$t = \frac{(\eta E / \delta)}{4\pi \lambda c_p} \left[\left(-\frac{1}{500 - T_0} \right)^2 - \left(\frac{1}{800 - T_0} \right)^2 \right] \quad (2)$$

Where η denotes the welding efficiency; δ represents the workpiece thickness (mm); λ is the thermal conductivity of the material (W/(m·K)); c_p signifies the specific heat capacity (J/(kg·K)); and T_0 indicates the initial temperature (K). As evidenced by Eq. (2), elevated E reduces post-weld cooling rates, prolonging the dwell time within the solidification temperature range. This thermal condition amplifies GB migration driving forces and promotes equiaxed grain coarsening in Mg alloy weld centers.

In addition, E determines the peak temperature in the region external to the molten pool ($T_{\max}$), with their interdependent relationship quantitatively described by Eq. (3):

$$T_{\max} = T_0 + \frac{0.234E}{c_p r_0^2} \quad (3)$$

Where r_0 corresponds to the distance from the observation point to the heat source. As derived from Eq. (3), increased E directly elevates the peak temperature in welded piece. Within the molten pool zone, excessive E values elevating the peak temperature beyond the boiling point of low-boiling-point elements in Mg alloys trigger metallic vapor jetting and keyhole instability. Conversely, in the HAZ, overly high peak temperatures coupled with energy input exceeding the activation energy barrier for grain boundary migration induce abnormal grain growth and boundary coarsening.

Table 1
Relationship between thickness and E as well as joint efficiency.

Model number	Thickness (mm)	Welding process	Heat input (E , J/mm)	Ultimate tensile strength (UTS, MPa)	Elongation (%)	Joint efficiency (%)	Ref.
AZ31	2.7	LBW	36	>BM	<BM	>100	[52]
AZ31	3	LBW	66.7	233.41	9.2	82.3	[53]
AZ31	4	LBW	83.3	238.5	13.9	<100	[54]
AZ31	5	LBW	76.6	238.5	12	96.1	[55]
AZ31	3	LBW	24	252	7.2	96.9	[56]
AZ31	5	LBW	62.5	227	11.2	94.6	[51]
AZ31	5.4	LB-CMT	266	252	4.49	93.3	[57]
AZ31	3.5	LBM	60	230.7	10.4	102.4	[58]
AZ80	10	LBM	2×10^4	284	9.2	92.0	[59]
ZK60	3.5	LBM	50	211.18	5.77	89.8	[58]
AZ31+0.4Ce	0.7	LBW	6.25	134	10.2	76.6	[60]
AM50+2Gd	1	LBW	11.2	224	/	97.0	[61]
AZ31+NZ30K	3.5	LBW	50	200.4	8.3	89.0	[62]
NZ30K	4	LBW	60	228	8.5	95.0	[21]
NZ30K	6	LBW	140	157	4.3	79.7	[63]
NZ30K	3.5	LBW	60	213.8	8	104.3	[58]
NZ30K	9.5	LB-TIG	/	212	/	103.4	[64]
NZ30K	9.5	LBM	233.3	193.55	7.69	99.3	[65]
NZ30K	9.5	LBM	150	198.56	6.68	101.7	[66]
ME21	1.4	LBW	43.6	181.3	/	80.1	[18]
WE43	10	TIG	566.7	195	5.1	87.8	[67]

This analysis confirms heat input as the predominant factor governing welded joint performance in Mg alloys. Process optimization requires precise heat input moderation to mitigate property degradation, yet must maintain sufficient energy density to ensure complete penetration. Wang et al. [51] demonstrated a linear correlation in 5-mm-thick AZ31B Mg alloy welds, where each 1 kW laser power increment increased penetration depth by 2~3 mm. As shown in Table 1, the relationship between the thickness of the welded sheet and the E as well as the joint efficiency is presented.

The parameters presented in Table 1 represent experimentally validated optimal values determined through systematic investigations. Empirical data confirm that these heat input levels reliably satisfy the fundamental process requirement of complete penetration. Crucially, the determination of these optimal parameters involves interdependent correlations among experimental conditions, welding process variables, and base metal (BM) chemical composition, rather than relying solely on sheet thickness. Consequently, the tabulated optimal heat input and corresponding sheet thicknesses do not exhibit a singular mathematical relationship. Nevertheless, welded joints inevitably exhibit performance degradation across varying BM compositions, heat input, and sheet thicknesses, fundamentally attributed to microstructural heterogeneity within the joint region [59,68, 69]. As an inherent characteristic of laser welding, microstructural evolution in the FZ remains unavoidable. Notwithstanding the development of solid-state joining techniques such as friction stir welding (FSW), these methods still fail to achieve complete microstructural consistency between weldments and base materials.

Therefore, the fundamental challenge in laser welding processes resides in achieving the dual objectives of ensuring

complete penetration while minimizing heat input to mitigate microstructural disparities. To this end, researchers have effectively reduced heat input while ensuring complete penetration by optimizing the process parameters of laser welding, adopting pulsed laser modulation technology [70], using ultrasonic vibration assistance [71–73], and developing new composite processes (such as laser-arc hybrid welding) [57,74, 75], etc.. Thereby inhibiting the microstructure heterogeneity in the joint area and improving the mechanical properties.

2.3. The influence of heat input on the microstructure evolution of the joint

Contemporary research on Mg alloy laser welding predominantly concentrates on deep penetration welding, with RE-free Mg alloys and systematic optimization of critical process parameters serving as the primary investigative focus. As the predominant determinant of joint quality, heat input exerts critical control over both microstructural characteristics (e.g., grain morphology, phase distribution) and mechanical performance metrics (e.g., ultimate tensile strength, fatigue resistance) in joints.

N. K. Babu et al. [56] systematically investigated AZ31 alloy joints, revealing that increasing the welding speed from 3 m/min to 5 m/min reduced FZ equiaxed grain size from 22 μm to 16 μm , concurrently elevating tensile strength from 237 MPa to 252 MPa, reaching 97 % of BM. This enhancement stems from reduced heat input and accelerated cooling rates, which suppress dendritic growth while promoting grain refinement through restricted GBs migration and enhanced constitutional supercooling effects. Complementary research by Shen et al. [76] on AZ61 alloy further validated these mechanisms. With the increase of welding speed, the

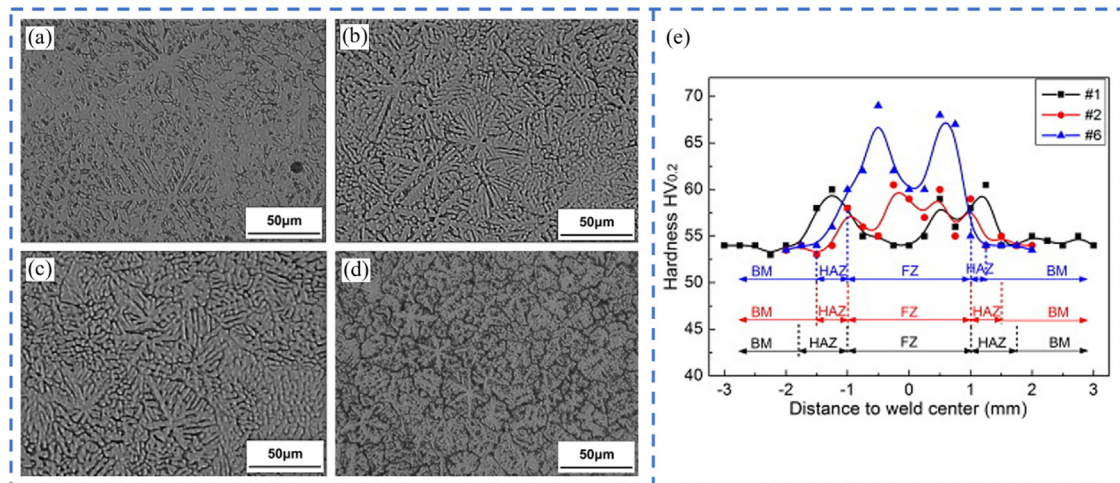


Fig. 3. Microstructure of the welds, (a) #1, (b) #2, (c) #4, (d) #6, and (e) hardness distribution maps of typical welds [51].

equiaxed refinement of α -Mg and the dispersed distribution of β -Mg₁₇Al₁₂ precipitated phase work together to achieve the simultaneous improvement of joint strength and elongation.

The distinct hexagonal close-packed (HCP) lattice characteristics of Mg and rapid solidification dynamics in laser welding synergistically govern FZ microstructural evolution. In the study of Wang et al. [51], the FZ grains presented a typical hexagonal equiaxed dendrite structure, and the dendrite growth showed a preferred orientation along the $\langle 11\bar{2}0 \rangle$ crystal direction. The research of Gao et al. [57] also observed the same phenomenon. As shown in Fig. 3 the heat input amounts of samples 1, 2, 4, and 6 are 85.7 J/mm, 62.5 J/mm, 44.0 J/mm, and 48.0 J/mm respectively. With the increase of heat input, the average half-axis length of dendrites shows a significant growth trend. Combined with the microhardness distribution measurements in Fig. 3(e) for Samples 1, 2, and 6 with welding speeds of 1.4 m/min, 2.4 m/min, and 7.8 m/min respectively, the FZ consistently demonstrates higher microhardness than the BM, with FZ microhardness progressively increasing with welding speed. This phenomenon can be attributed to the fact that the low heat input promotes the formation of a finer microstructure in FZ, the grain refinement effect promotes the increase of microhardness, and this strengthening effect is significantly better than that of the base material matrix.

During the non-equilibrium solidification process, dendrite morphology is primarily dominated by local constitutional undercooling. In the AZ31 alloy employed by Wang et al. [51], Al chiefly restrains dendritic growth via pinning by second-phase particles at GBs, having a minimal effect on the interfacial energy anisotropy [77]. In contrast, Zn directly modulates interfacial energy anisotropy parameters. Fig. 4 presents the 3D reconstructed dendritic structures in Mg-Zn alloys [78]. At $\text{Zn} \leq 20$ wt.%, dendrites exhibit bidirectional growth along $\langle 11\bar{2}0 \rangle$ and $\langle 11\bar{2}\bar{3} \rangle$. At $\text{Zn} \geq 45$ wt.%, $\langle 11\bar{2}0 \rangle$ growth is inhibited, forming 12-branch morphologies. Within the transitional range of 20 wt.% to 45 wt.%, highly branched, frac-

tal seaweed-like dendrites emerge, demonstrating orientation competition and growth instability.

Cross-correlating Figs. 3 and 4 reveals that despite the nominal 10 wt.% Zn content in Wang et al.'s alloy [51], its dendritic morphologies—particularly in Fig. 3(b-c)—encompass nearly the entire Zn-concentration spectrum. This phenomenon originates not merely from thermal input modulation of grain growth, but is primarily attributable to local segregation induced by rapid solidification; localized Zn enrichment induces the morphological transformation. This demonstrates that cooling rates amplify solute segregation's governing influence on interfacial energy anisotropy by modulating the degree of non-equilibrium solidification.

Heat input exerts a significant influence on both the width of the partially melted zone (PMZ) and grain characteristics in the HAZ. Wang et al. [51], demonstrated that elevated heat input prolongs high-temperature retention in molten pools, causing PMZ width expansion from a narrow fusion line interface to several grain widths. Concurrently, excessive heat input induces HAZ grain coarsening and β -Mg₁₇Al₁₂ phase coarsening. Liu et al. [69] investigated laser welding of T4-treated AZ80 alloy under a heat input of 2×10^4 J/mm. The FZ comprised coarse equiaxed grains and columnar grains with average sizes of 38.8 μm and 23.2 μm respectively, representing 5.4 times and 3.2 times the grain size of the BM. Notably, the HAZ demonstrated pronounced GBs coarsening. As depicted in Fig. 5, β -Mg₁₇Al₁₂ precipitates were identified in the microstructure. The dislocation density in the FZ surged from $(0.5 \sim 0.9) \times 10^{14} \text{ m}^{-2}$ in the BM to $(8.4 \sim 13.3) \times 10^{14} \text{ m}^{-2}$, representing a 15.8-fold enhancement. From the perspective of mechanical performance, the welded joint retained 92 % of the BM's tensile strength through the synergistic effects of dislocation density augmentation and FZ precipitation strengthening, despite undergoing substantial grain coarsening. Notably, while β -phase precipitation partially offset strength degradation, the joint elongation plummeted from 13.02 % (BM) to 4.49 %, indicating a 65.5 % ductility reduction.

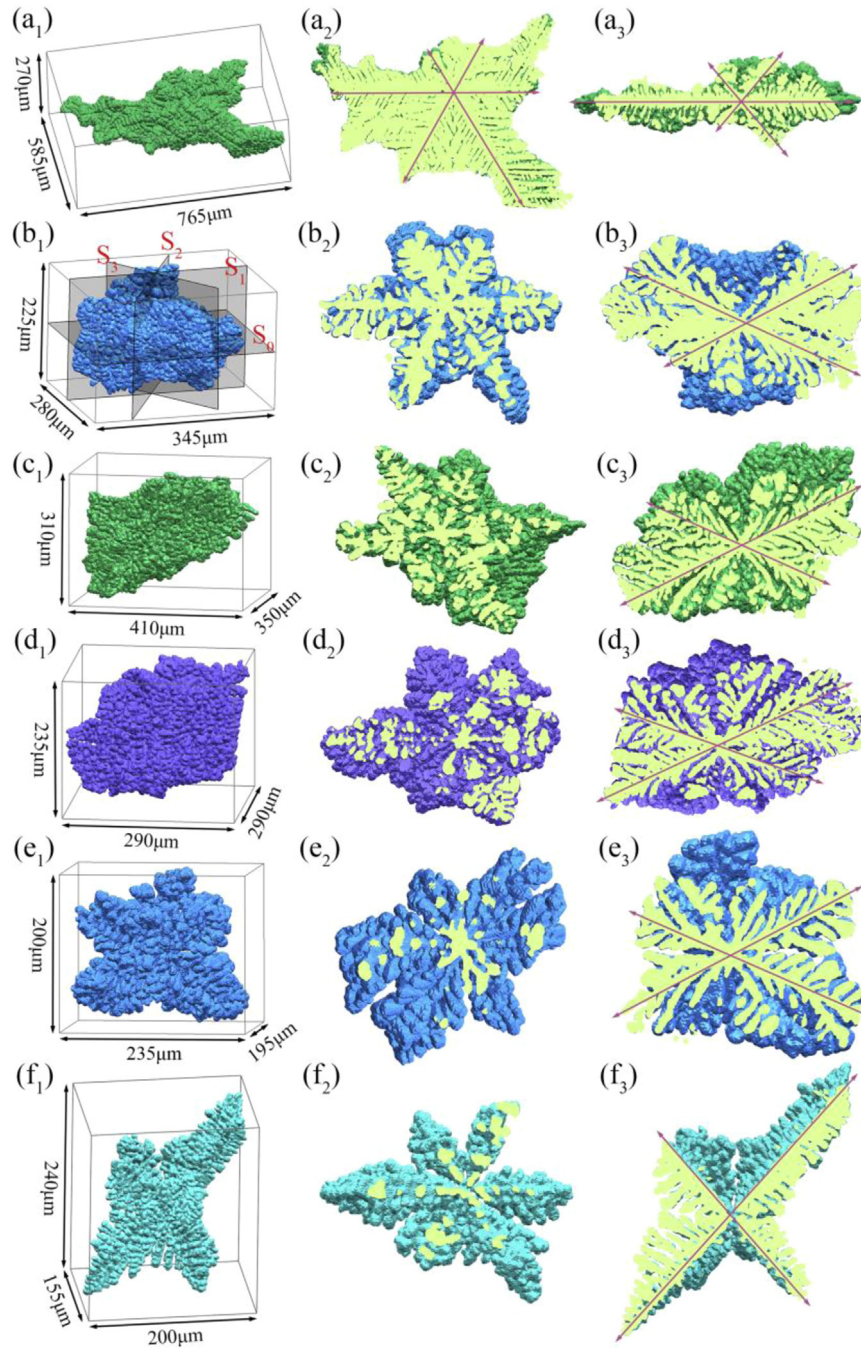


Fig. 4. The 3D reconstructed dendrite morphology of Mg-(a₁-a₃) 20 wt.%, (b₁-b₃) 25 wt.%, (c₁-c₃) 30 wt.%, (d₁-d₃) 35 wt.%, (e₁-e₃) 40 wt.% and (f₁-f₃) 45 wt.% Zn, respectively. The second and third columns show the dendrite morphologies viewed from $\langle 0001 \rangle$ and $\langle 10\bar{1}0 \rangle$ directions, respectively [78].

In the aforementioned study, the observed dislocation multiplication fundamentally arises from the interplay between residual stress during the post-solidification cooling stage and its induced localized plastic strain. Upon solidification completion, thermal gradients cause differential cooling contraction across material regions, thereby generating residual stresses. When these stresses exceed the material's yield strength at the prevailing temperature, localized plastic strain is activated, which directly promotes extensive dislocation multiplication through mechanisms such as Frank-

Read sources. This thermally triggered progressive interplay of stress-strain responses—termed thermo-mechanical coupling—ultimately leads to intensive dislocation proliferation.

These studies demonstrate that the strengthening-toughening process in welded joints originates from the synergistic coupling between rapid solidification-induced microsegregation and thermo-mechanical effects, both of which modulate interactions among competing strengthening mechanisms through microstructural mediation. Specifically, local

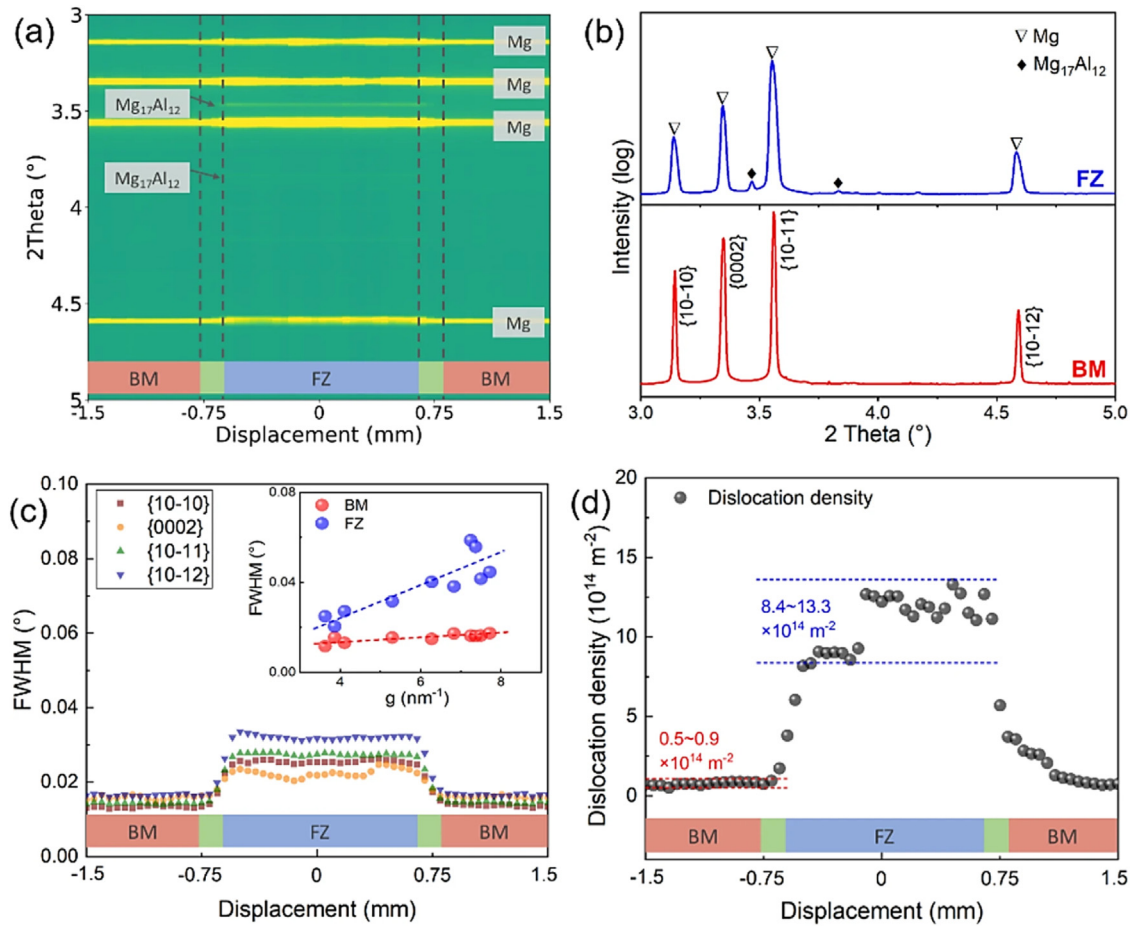


Fig. 5. Synchrotron X-ray diffraction analysis in the welding joint: (a) diffraction heat map around a welding joint, (b) integrated diffraction profiles of BM, and FZ, (c) FWHM evolution of four selected Mg peaks, insert is the Williamson-Hall (WH) plots for Mg peaks at BM, and FZ, and (d) dislocation densities determined by the convolutional multiple whole profile (CMWP) method [59].

solute segregation governs dendritic evolution by altering solid-liquid interfacial energy anisotropy, indirectly influencing grain refinement. This refinement concurrently enhances strength and toughness via the Hall-Petch mechanism. Precipitation strengthening operates through second phase particles obstructing dislocation motion, while dislocation strengthening arises from strain-field interactions (e.g., intersection, entanglement) that impede subsequent dislocation glide. Critically, these four mechanisms do not operate in isolation but exhibit pronounced concurrent competition and synergy, with the equilibrium of their interactions critically dictating the joint's ultimate performance. These discoveries establish critical theoretical foundations for subsequent microstructural regulation through post-weld treatments, enabling precision optimization of welded joint properties [79–82].

In-depth investigations into welding mechanisms reveal distinct research priorities across Mg alloy systems. For RE-free Mg alloys, welding studies predominantly focus on parameter optimization and process stability control, where heat input emerges as the pivotal parameter governing microstructural evolution and performance. Reducing thermal input facilitates grain refinement, curtails dendritic growth, and mitigates both solute segregation and metastable phase

retention. This approach simultaneously shortens the molten pool's high-temperature dwell time, effectively constraining HAZ grain coarsening and suppressing PMZ width expansion.

In contrast, the superior properties of Mg-RE series alloys primarily rely on the grain refinement effect of RE elements and the precipitation strengthening effect. Consequently, welding research for this alloy system emphasizes RE elements' evolutionary behavior during welding and their critical influence on joint performance. As evidenced in Table 1, Mg-RE welded joints exhibit significantly higher property degradation than RE-free Mg counterparts, a phenomenon intrinsically linked to compromised RE second phases regulation and severe solute atom segregation induced by rapid solidification of the molten pool.

3. The role of RE elements in the welding of Mg-RE alloys

3.1. The impact of RE on the efficiency of welded joints

RE elements serve as crucial strengthening components in magnesium alloys. Their strengthening-toughening mech-

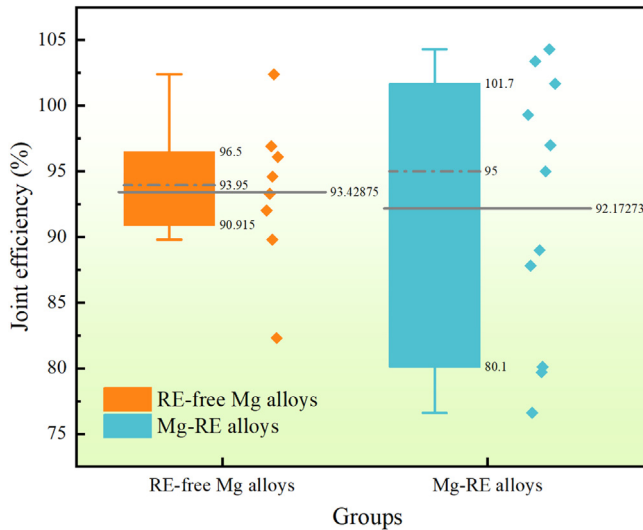


Fig. 6. Box plot of joint efficiency.

anism under conventional casting or wrought conditions are supported by a well-established theoretical framework [16,83–85]. However, the inherent non-equilibrium solidification characteristics of laser welding fundamentally alter the dynamic behavior pathways of RE elements, resulting in unique performance characteristics of the welded joints that diverge from conventional understanding.

The mechanical properties of welded joints depend not only on process parameters but are also closely related to BM composition. Therefore, direct comparison of mechanical properties between joints of different alloys cannot objectively evaluate their process compatibility or weldability. To eliminate the influence of the BM's intrinsic strength and enable normalized assessment, the joint efficiency—calculated as $(UTS_{\text{welded joints}}/UTS_{\text{BM}}) \times 100\%$ [56–59]—is adopted as the core evaluation metric.

Fig. 6 presents a statistical analysis of joint efficiency for Mg-RE alloys and RE-free Mg alloys, with data sourced from Table 1. The yellow box plot and yellow diamond markers represent the statistical results and raw data for RE-free Mg alloys, respectively, while the blue box plot and blue diamond markers correspond to Mg-RE alloys. The dotted and solid lines in the plot indicate the median and mean values of the data, respectively. Analysis reveals significant differences in the performance distribution characteristics between the two alloy systems. The data points for RE-free Mg alloys are predominantly concentrated in the high-strength, high-efficiency region, demonstrating exceptional performance stability. In contrast, Mg-RE alloys exhibit substantially greater dispersion, indicating more heterogeneous performance characteristics.

A comparative assessment of the two alloy systems shows that, Mg-RE alloys exhibit two distinct characteristics: (1) their joint efficiency is generally lower within conventional welding parameter ranges; (2) the performance data show high dispersion, indicating pronounced sensitivity to welding parameters, heat input, and pre-/post-processing conditions.

This performance sensitization and instability fundamentally stem from the altered strengthening mechanisms of RE elements induced by the non-equilibrium solidification of laser welding. In near-equilibrium conventional processing, RE elements primarily strengthen the matrix through equilibrium-phase precipitates. Under laser welding conditions, however, the extremely high cooling rates significantly suppress the precipitation of such equilibrium phases. Meanwhile, the welding thermal cycle also triggers dissolution, coarsening, or transformation of pre-existing second phases in the BM. Consequently, the conventional precipitation strengthening effect is substantially diminished, while new strengthening structures fail to form stably, collectively leading to macroscopic performance instability.

3.2. Laser welding of non-equilibrium solidification and RE behavior

Unlike near-equilibrium solidification in conventional casting, laser welding is a complex physical metallurgical process involving transient localized heating, melting, and rapid solidification, exhibiting pronounced non-equilibrium characteristics [2,23, 86]. While this extreme environment suppresses conventional precipitation strengthening mechanisms involving RE elements, it creates unique opportunities for them to enhance weld performance through GBs segregation, grain refinement, and suppression of welding defects.

During the heating stage, laser welding differs markedly from conventional casting. Laser welding employs a high-energy-density heat source ($10^6 \sim 10^7$ W/cm²) with an extremely rapid heating rate, causing instantaneous material melting and formation of a molten pool [43,44, 87]. This intense heating promotes vaporization of low-boiling-point elements, increasing the risk of welding defects, while initial secondary phases dissolve rapidly without sufficient time for long-range diffusion of alloying elements [88]. In contrast, conventional casting involves gradual heating that permits adequate elemental diffusion and more complete dissolution of secondary phases. Consequently, the initial distribution of RE elements in laser welding exhibits pronounced heterogeneity, establishing a critical precondition for subsequent solute redistribution within the molten pool.

During the melting stage, the internal flow, temperature distribution, and stability of the molten pool differ fundamentally between the two techniques. As mentioned in Section 2.1, in the deep penetration welding mode of laser welding, a deep and narrow keyhole molten pool is formed [44]. At this time, there is a strong Marangoni convection driven by the surface tension gradient and an opposing pressure generated by the metal vapor. The molten pool has a short existence time and a large temperature gradient [87,89, 90]. The vigorous fluid flow facilitates macroscopic solute transport but also intensifies pool instability, which significantly influences bubble escape and pore distribution [91] (Section 4). Conversely, conventional casting employs a large melt volume with slow cooling, yielding a uniform temperature field and sufficient time for homogeneous solute distribution. In the laser weld-

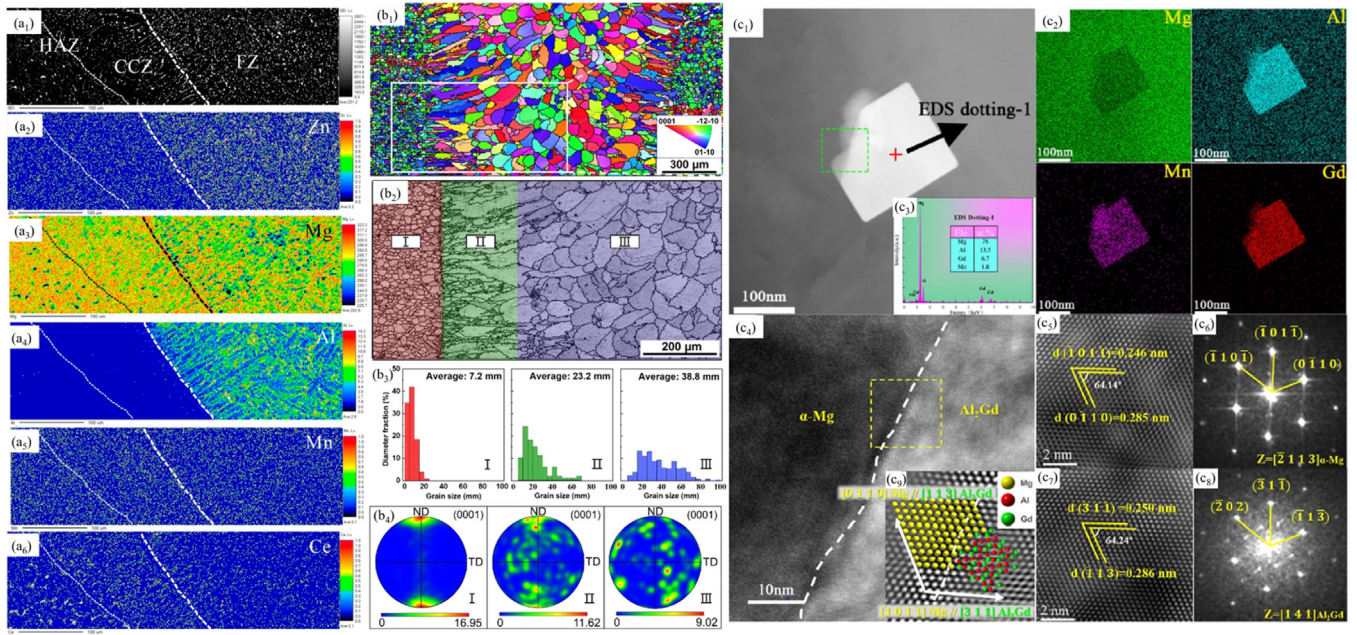


Fig. 7. Core effects in molten pool solidification: (a₁–a₆) solute trapping [18]; (b₁–b₄) interface instability [59]; (c₁–c₈) metastable phase formation [61].

ing process, however, RE elements exhibit relative enrichment in the pool due to their high boiling points which resist vaporization. This very turbulence then facilitates a uniform distribution of these elements, setting the stage for heterogeneous nucleation upon solidification.

Finally, at the solidification stage, the differences in microstructure formation mechanisms are most critical. The characteristically high specific heat capacity and thermal diffusivity inherent to magnesium alloys further intensify the fundamental "rapid heating and rapid cooling" nature of laser welding. This amplification produces even steeper thermal gradients within the molten pool and accelerates the solidification process. Consequently, laser welding of magnesium alloys manifests as extreme non-equilibrium solidification, giving rise to three core effects. (1) Solute Trapping: The solid-liquid interface advances at a velocity far exceeding the solute diffusion rate, causing solute atoms such as RE elements to be captured by the rapidly solidifying interface. This leads to severe segregation at GBs [18,92, 93], as shown in Fig. 7(a₁–a₆). (2) Interface Instability: The extremely high thermal gradients and cooling rates reduce the stability of the solid-liquid interface, promoting the formation of fine cellular crystals, columnar crystals, or equiaxed dendritic structures in the FZ [56,59, 94], as depicted in Fig. 7(b₁–b₄). (3) Metastable Phase Formation: The solidification process significantly deviates from the equilibrium phase diagram, readily resulting in the formation of supersaturated solid solution and even directly promoting the precipitation of non-equilibrium metastable strengthening phases [61,95, 96]. Fig. 7(c₁–c₈) presents TEM analysis of the Al₂Gd phase in the weld metal [61]. The images clearly show a coherent interface between the Al₂Gd phase and the α -Mg matrix. This phase constitutes a typical metastable phase and serves as a crucial strengthening source in the weld. This result unequivocally demonstrates that the rapid solid-

ification in laser welding alters both the types and formation pathways of strengthening phases, shifting from coarse rare-earth equilibrium phases toward finer, less stable metastable phases. The effects of solute trapping and interface instability during solidification have been extensively discussed in Sections 2.2 and 2.3, respectively, and will not be reiterated here.

The uniqueness of laser welding for rare-earth magnesium alloys manifests not only in the rapid solidification process of the molten pool but also extends to the microstructural evolution within the HAZ. The welding heat input serves as the key external variable that modulates the extent of this non-equilibrium solidification and thereby governs the evolutionary behavior of RE elements. As indicated in Eqs. (2) and (3), the heat input (E) directly determines the molten pool's peak temperature, its retention time at elevated temperatures, the cooling rate, and the peak temperature of the thermal cycle.

Excessive heat input prolongs the retention time at high temperatures. While this may facilitate solute diffusion, it simultaneously exacerbates elemental evaporation loss and GBs segregation, and may induce either excessive dissolution or coarsening of pre-existing strengthening phases within the HAZ [67,95]. Conversely, insufficient heat input implies extremely rapid cooling rates. This condition intensifies solute trapping while potentially causing severe compositional heterogeneity and stress concentration due to the accelerated solidification [92,93].

It is evident that laser welding profoundly influences the properties of Mg-RE alloys. A complex coupling relationship exists between the welding heat input and the behavior of RE elements. The heat input directly governs the intensity of non-equilibrium solidification and critically determines the final manifestation of RE elements' effects—whether they tend

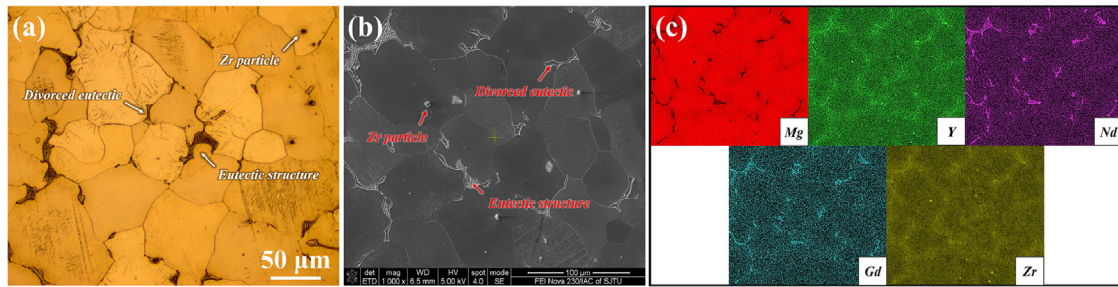


Fig. 8. (a) Optical and (b) SEM micrographs of low-pressure sand-cast alloy. (c) Corresponding EDS maps of SEM micrograph in (b) [67].

toward beneficial grain refinement and secondary phase formation, or transform into detrimental GBs segregation and supersaturated solid solutions. This process directly dictates the final performance of the weld joint. Consequently, precisely understanding the behavioral patterns of RE elements during laser welding is crucial for achieving controlled performance in Mg-RE alloy welded joints. Based on this understanding, subsequent sections will systematically analyze the dynamic evolution laws of secondary phases under laser welding conditions (Section 3.3) and the comprehensive strengthening-toughening mechanisms of RE elements (Section 3.4).

3.3. The influence of heat input on the Mg-RE second phase

Thermal input stands as a critical determinant of joint microstructural evolution. RE elements exhibit intensified grain boundary segregation with increasing heat input, while precipitate formation becomes markedly suppressed due to elemental partitioning. In the research of Tong et al. [67], when welding current (I) reached $I \geq 190$ A, continuous network-like eutectic phases enriched with RE elements (Y, Nd, Gd) and Zr formed in the HAZ, as illustrated in Fig. 8. This segregation phenomenon stems from high thermal input conditions. Elevated heat input subjects the HAZ to peak temperatures approaching the solidus line and generates sharp thermal gradients. Concurrently, the elevated temperature significantly enhances the diffusion capacity of solute atoms, while the steep thermal gradients combined with rapid cooling jointly drive solute enrichment at the solidification front. Ultimately, the enriched solute forms a continuous network-like eutectic phase along GBs via non-equilibrium solidification.

Fig. 9(a-b) displays the XRD patterns of the FZ and BM, indicating no new phases formed during the welding process. The α -Mg diffraction peaks in the FZ exhibit a non-monotonic shift—first to lower angles and then to higher angles—with increasing welding current. At 190 A, the diffraction angle reaches its minimum value. In accordance with Bragg's law, it suggests enhanced solid solubility of RE elements in the Mg matrix and subsequent lattice distortion. When the current further rises to 210 A, the rightward peak shift reflects the onset of RE elements precipitation from the matrix. Differential scanning calorimetry (DSC) analysis in Fig. 9(c) reveals distinct thermal responses: except for the $I = 210$ A joint and BM, other FZ exhibit no discernible

eutectic decomposition endothermic Peak B at 540 °C, but instead display broadened Peak A within 350~450 °C. This indicates that under medium heat input, RE elements exist in a supersaturated solid solution state, failing to form effective strengthening phases. Conversely, high heat input (210 A) promotes partial precipitation of RE elements, though without achieving complete desorption. Notably, during welding at 170 A, despite the low segregation degree of RE elements, the joint strength only recovers to 87.8 % of the BM due to insufficient precipitated strengthening phases. This phenomenon confirms that in Mg alloy welding, heat input modulates joint performance by regulating the dynamic equilibrium between RE elements solid solution and precipitation.

In the research of laser welding Mg-8Gd-3Y-Zr under T6 condition by Wang et al. [97], the joint strength of 317 MPa was primarily attributed to ultrafine-grained structures in the FZ and the constraining effect of RE precipitates on grain coarsening in the HAZ. Their subsequent research systematically investigated the high-temperature performance of laser-welded joints from ambient to 300 °C [98]. Following post-weld heat treatment (PWHT) involving 430 °C solution treatment and 225 °C aging, the joints exhibited significantly enhanced elevated-temperature strength markedly due to the precipitation of β' phase in both the FZ and HAZ, retaining a tensile strength of 158 MPa at 300 °C.

In subsequent research, Wang et al. [95] delved deeper into the dynamic evolution patterns of HAZ precipitates and their influence on mechanical properties during laser welding of T6-treated Mg-8Gd-3Y-Zr alloy. As shown in Fig. 10(a), researchers have divided HAZ into three typical regions: the complete dissolution region (CDZ, region I), the partial dissolution and transformation region (PDTZ, Region II), and the substrate proximity region (Region III). Fig. 10(b) presents the microhardness gradient across the HAZ. Fig. 11(a₁-a₂) demonstrates that within Region I adjacent to the FZ, β' strengthening precipitates were completely dissolved due to prolonged high-temperature exposure. Solute atoms (Gd, Y) diffused into the matrix, while rapid cooling during laser welding suppressed reprecipitation. As demonstrated in Fig. 11(b₁-c₂), partial dissolution of β' phase occurs in Zone II, accompanied by the formation of non-equilibrium precipitates. Observed along the $[2\bar{1}\bar{1}0]_{\alpha}$ zone axis, these newly formed phases exhibit elliptical and elongated rod-like morphologies. Originating from localized

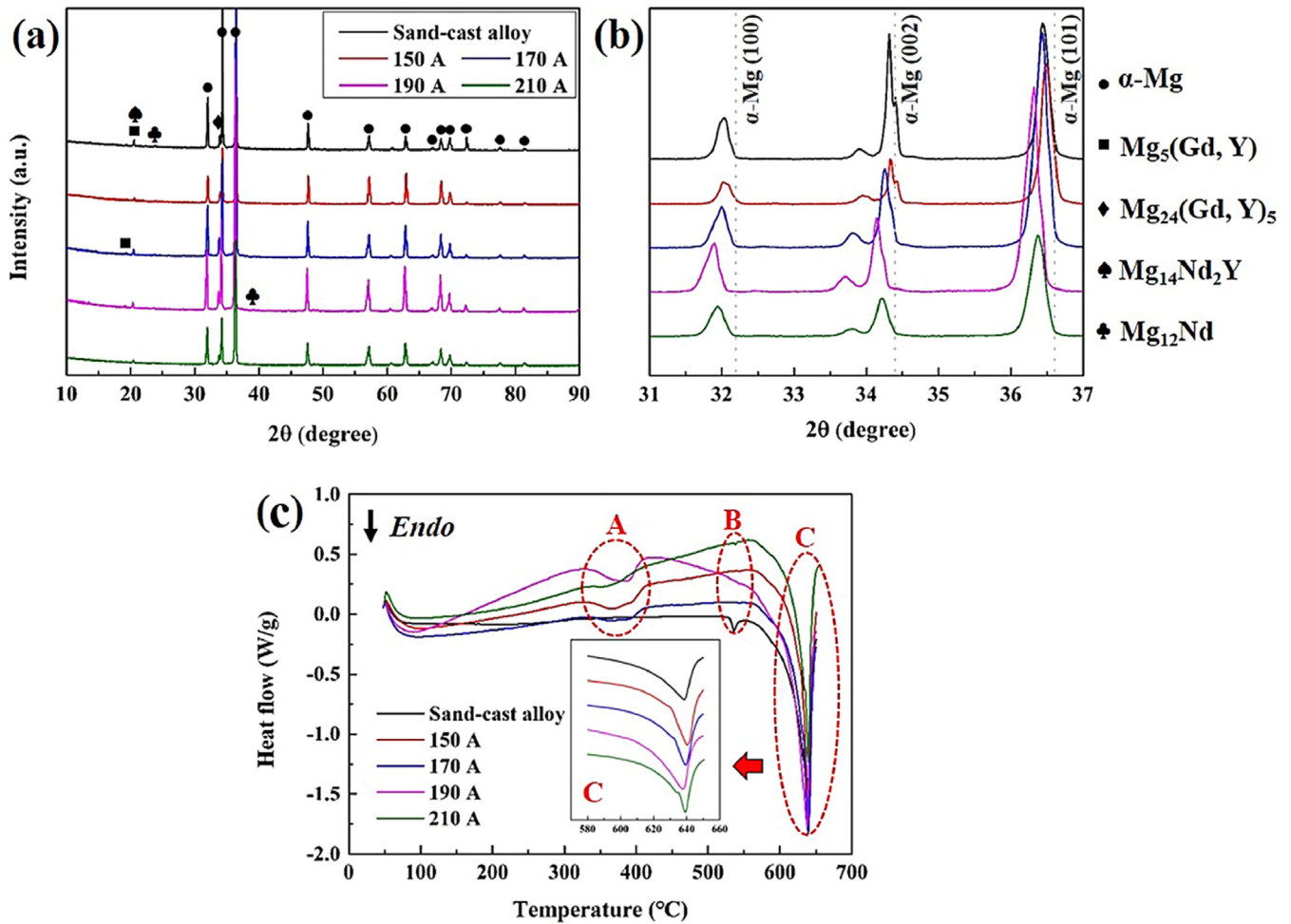


Fig. 9. (a) XRD spectra of the BM and FZ in the joints with different welding currents. (b) XRD spectra with the scanning angle from 31° to 37° in (a). (c) DSC analysis of the FZ in the joints repaired with different welding currents [67].

solute enrichment under rapid welding thermal cycles, their distinct morphology and crystallographic orientation deviate markedly from conventional aging sequences, unequivocally demonstrating the predominance of non-equilibrium phase transformation mechanisms. Within Zone III, β' precipitates exhibited an over-aged state [99], accompanied by the nucleation of acicular β -phase particles as shown in Fig. 11(d₁-d₂). The formation of β -phase was constrained by the transient thermal cycles during welding, with nucleation limited to solute-enriched regions and displaying heterogeneous spatial distribution characteristics [100].

The microhardness gradient across the HAZ in Fig. 8(b) reveals three distinct thermomechanical states: (1) The CDZ acts as a mechanical weak zone due to complete β' dissolution; (2) The PDTZ partially recovers hardness through non-equilibrium precipitation; (3) Zone III retains residual strengthening effects from over-aged β' precipitates. Crucially, the rapid heating/cooling kinetics inherent to welding thermal cycles severely compressed the diffusion-controlled timeframe for the $\beta' \rightarrow \beta$ phase transformation, ultimately leading to a highly heterogeneous precipitate distribution.

This analysis reveals that the rapid heating/cooling characteristics inherent to welding thermal cycles modulate the dissolution, reprecipitation, and phase transformation behaviors of precipitates in the HAZ, thereby governing the microstructural evolution and mechanical property gradients across this region. Crucially, thermal cycle parameters—including peak temperature, cooling rate, and cycle frequency—exert dual regulatory effects on microstructural development. Specifically, transient high-temperature exposure restricts dynamic phase transformation processes in the HAZ, resulting in reduced volume fractions of β' strengthening phases, which constitutes the predominant mechanism underlying joint hardness degradation. Conversely, excessive heat input induces detrimental coarsening of HAZ precipitates and subsequent weakening of strengthening effects.

The performance of Mg-RE alloy welded joints is primarily governed by the dynamic evolution of RE elements across distinct weld regions. Beyond the HAZ, heterogeneous thermal fields and complex thermal cycling characteristics induce differentiated evolution in two critical zones: (1) rapid solidification in the FZ exacerbates intergranular solute segregation and promotes preferential nucleation of micro-defects

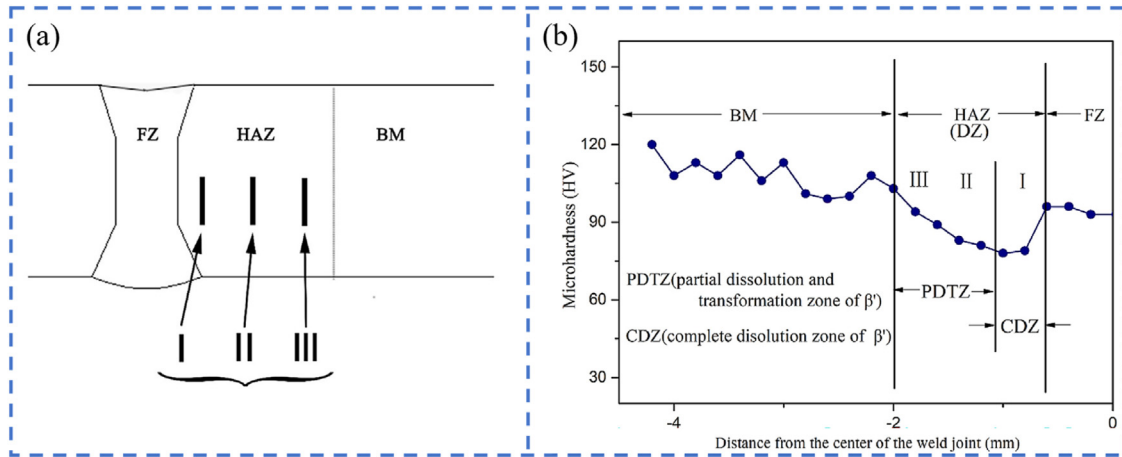


Fig. 10. (a) Schematic representation of different zone of the laser welded joint, (b) a typical microhardness profile across half a welded joint along mid-thickness line [95].

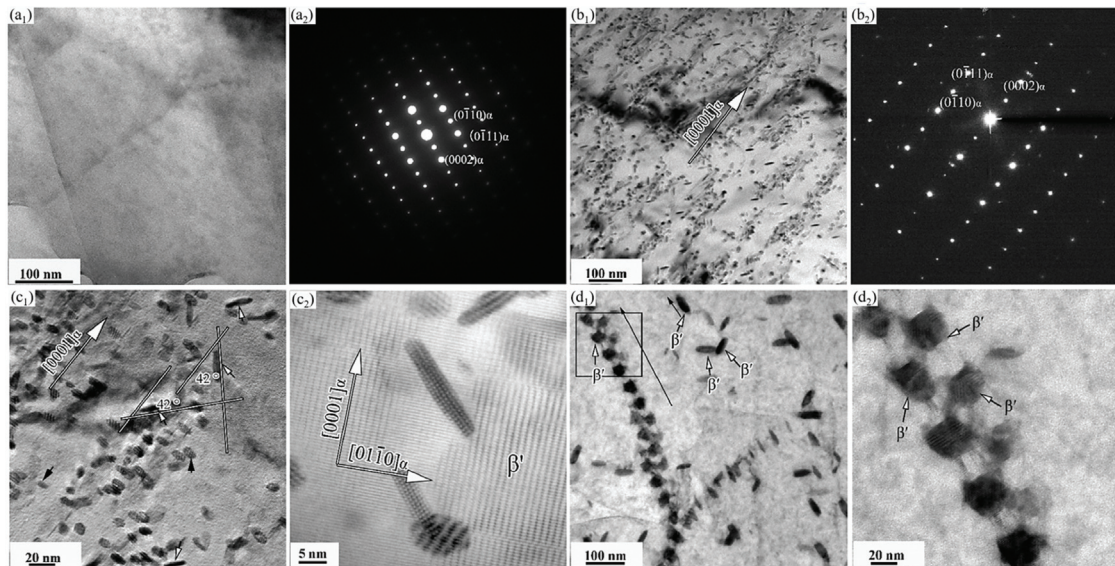


Fig. 11. TEM images recorded from the HAZ close to FZ (I, II) along the zone axis $[2\bar{1}\bar{1}0]_{\alpha}$ and (III) along the zone axis $[0001]_{\alpha}$. Zone I: (a₁) BF image; (a₂) the corresponding SAED pattern of the precipitates. Zone II: (b₁) BF image; (b₂) the corresponding SAED pattern; (c₁) the enlarged BF image showing the morphology of precipitates; (c₂) the enlarged image of typical long rod phase. Zone III: (d₁) BF image; (d₂) the enlarged BF image showing the morphology of precipitates from the black rectangular region [95].

[101]; (2) the PMZ exhibits altered spatial distribution of second phases due to coupled thermo-mechanical and phase transformation stresses [102–105]. Notably, the gradient distribution of precipitates in the HAZ and the microscale segregation patterns of RE elements in the FZ collectively regulate grain growth orientation and competitive mechanisms, thereby dictating the microstructural heterogeneity of the weld. This multi-factor interaction ultimately dictates the residual stress distribution, mechanical property variations, and high-temperature stability of the welded joint.

Therefore, optimization strategies should integrate welding process control with material design, including: employing pulsed laser techniques to regulate heat input for promoting metastable phase precipitation [57,106–108]; introducing external field disturbances (e.g., electromagnetic stirring) to enhance molten pool fluidity and refine solidified microstruc-

tures; incorporating heterogeneous nucleation particles via microalloying to achieve grain refinement [19,20, 67,96, 109–111]; and designing graded aging treatments to optimize precipitate distributions [59,97, 98,112]. These approaches collectively improve the strength-toughness balance and stability of welded joints through multiscale microstructural regulation.

3.4. Microstructure regulation and strengthening mechanism of RE

3.4.1. The mechanism of grain refinement induced by RE

Under rapid solidification conditions during laser welding, RE elements play a decisive role in grain refinement within FZ by simultaneously increasing the degree of constitutional undercooling and providing heterogeneous nucleation sites. During high-energy beam welding, Mg (boiling

point 1090 °C) evaporates preferentially, while RE elements (Y, b.p. 3345 °C; Nd, b.p. 3100 °C; Gd, b.p. 3250 °C) [113] remain due to their higher boiling points, leading to their relative enrichment in the molten pool. Moreover, as detailed in Section 2.3, rapid solidification of the molten pool induces solute enrichment ahead of the solid-liquid interface. Therefore, the enrichment of RE elements intensifies constitutional undercooling and concurrently acts as heterogeneous nucleation sites, which suppresses the directional growth of α -Mg grains and promotes the formation of equiaxed grains within the fusion zone, leading to further refinement of the grain structure [114]. This grain refinement enhances mechanical properties via the Hall-Petch effect.

The combined effect of grain refinement and microstructural homogeneity elevates joint performance. Lei et al. [101] observed a 27 % hardness increase compared to the BM in laser-welded WE43 alloy, attributed to the synergistic effect of RE elements (Y, Nd) and rapid solidification. Tong et al. [67] demonstrated that reducing the weld grain size from 20 μm to 13 μm increased hardness to 77 HV_{0.3} (111.6 % of BM), TS from 178 MPa to 195 MPa, and EL to 124.4 % of BM. This enhancement stems from increased grain boundary density hindering crack propagation. Within their study, accelerated cooling and Zr-induced grain refinement were dominant factors, with Zr contributing significantly to refinement strengthening in the weld region.

Although Zr is not a RE element, it serves as a crucial grain refiner and remains one of the most frequently utilized alloying elements in the development of RE-containing Mg alloys. In Mg alloys, the grain refinement effect is predominantly achieved through the synergistic action of undissolved Zr particles (Zr_p) and dissolved Zr solute (Zr_s) [115–118]. Zr_p facilitates grain refinement by providing heterogeneous nucleation sites, whereas Zr_s suppresses grain growth via solute segregation-induced constitutional undercooling. Notably, Zr_s contributes more significantly to grain inhibition [119–121]. When TIG welding is applied to Mg alloys, excessive grain coarsening in the FZ and HAZ often occurs due to relatively high heat input [122]. Intriguingly, in the research conducted by Tong et al. [67] on a Zr-containing Mg-Y-RE-Zr alloy, the grain-refining effect of Zr substantially mitigated the adverse impacts of heat input and enhanced the tolerance window for welding parameters.

This analysis reveals that the grain refinement effect of RE elements originates not only from intrinsic mechanisms during solidification—including constitutional undercooling and heterogeneous nucleation—but also exhibits strong dependence on welding process parameters. The coupling effect between heat input and RE elements collectively governs the evolution of grain size and microstructural homogeneity in weldments [123,124]. Consequently, during welding operations, strategic optimization of heat input should be implemented to accommodate the transient, high-gradient characteristics of the molten pool. Specifically, prolonging the high-temperature residence time of the molten pool and incorporating external field interventions—such as electromagnetic stirring, ultrasonic vibration, and pulsed arc techniques—can enhance

Zr_p dissolution, accelerate $\text{Zr}_\text{s}/\text{Zr}_\text{p}$ diffusion, and promote uniform RE distribution [118–121,125, 126]. These synergistic approaches effectively optimize the refinement efficacy.

3.4.2. GBs pinning and second phase strengthening

In metal welding processes, grain coarsening within HAZ is a critical factor compromising joint performance. To address this, RE elements provide an innovative strategy for suppressing abnormal grain growth by forming thermally stable second-phase particles and generating substantial GBs pinning effects.

Chen et al. [19,20] conducted laser welding research on AZ31 alloy containing 0.4 wt.% Ce. In the BM, Ce forms high-melting-point RE compounds ($\text{Al}_{17}\text{Ce}_2\text{Mn}_7$, $\text{Al}_{11}\text{Ce}_3$) with Al and Mn, while Zn and Mg generate conventional low-melting-point compounds ($\text{Mg}_{17}\text{Al}_{12}$, MgZn_2). As shown in Fig. 12(a, b), remelting and solidification in the FZ produced dendritic structures surrounded by uniformly distributed fine precipitates comprising micrometer-sized RE particles ($\text{Al}_{17}\text{Ce}_2\text{Mn}_7$, $\text{Al}_{11}\text{Ce}_3$) and conventional particles ($\text{Mg}_{17}\text{Al}_{12}$, MgZn_2). These precipitates impede dislocation movement through the pinning effect, remarkably strengthening the FZ.

As observed in Fig. 12(c, d), spherical or rounded island-shaped second phases with coral-like internal morphology are detected. Based on EDS analysis, these precipitates are identified as eutectic products of $\text{Mg}_{17}\text{Al}_{12}$ and MgZn_2 . This phenomenon is attributed to the partial melting of the low-melting-point conventional compound $\text{Mg}_{17}\text{Al}_{12}$ under weld thermal cycles [127,128]. Driven by liquid-phase surface tension and spatially constrained by GBs, the molten $\text{Mg}_{17}\text{Al}_{12}$ coalesces with semi-molten lamellar MgZn_2 at GBs. This observation confirms pronounced thermal cycle effects on the HAZ. Remarkably, no distinct coarse grains are observed in the HAZ of the welded joint. This phenomenon is ascribed to the high-melting-point characteristics of RE compounds and their GBs pinning effects [129], which collectively suppress HAZ grain coarsening. Consistent observations were reported by Yin et al. [60], where solution treatment further verified that the refined RE particles—with average sizes below 1 μm and limited volumetric fraction—served as the dominant factor in restraining HAZ grain coarsening.

As a critical grain refiner, Zr can also effectively nail GBs. Hou et al. [110] conducted a comparative study on the microstructure and mechanical properties of Mg-6Gd-3Y and Mg-6Gd-3Y-0.5Zr filler wires during weld repair of Mg-6Gd-3Y-0.5Zr BM. As illustrated in Fig. 13, Zr-bearing precipitates were identified at GBs in welds fabricated with Zr-containing filler wire. These precipitates impede GBs migration in the FZ, thereby enabling the FZ to exhibit enhanced thermal stability during 6 h heat treatment at 500 °C, effectively suppressing grain coarsening.

In conclusion, RE elements demonstrate pronounced GBs pinning effects during welding through the formation of high-melting-point eutectic phases. On the one hand, the resultant nanoscale second phases effectively inhibit HAZ grain coarsening, thereby enhancing joint strength and toughness. On the other hand, particle-dislocation interactions strengthen the FZ

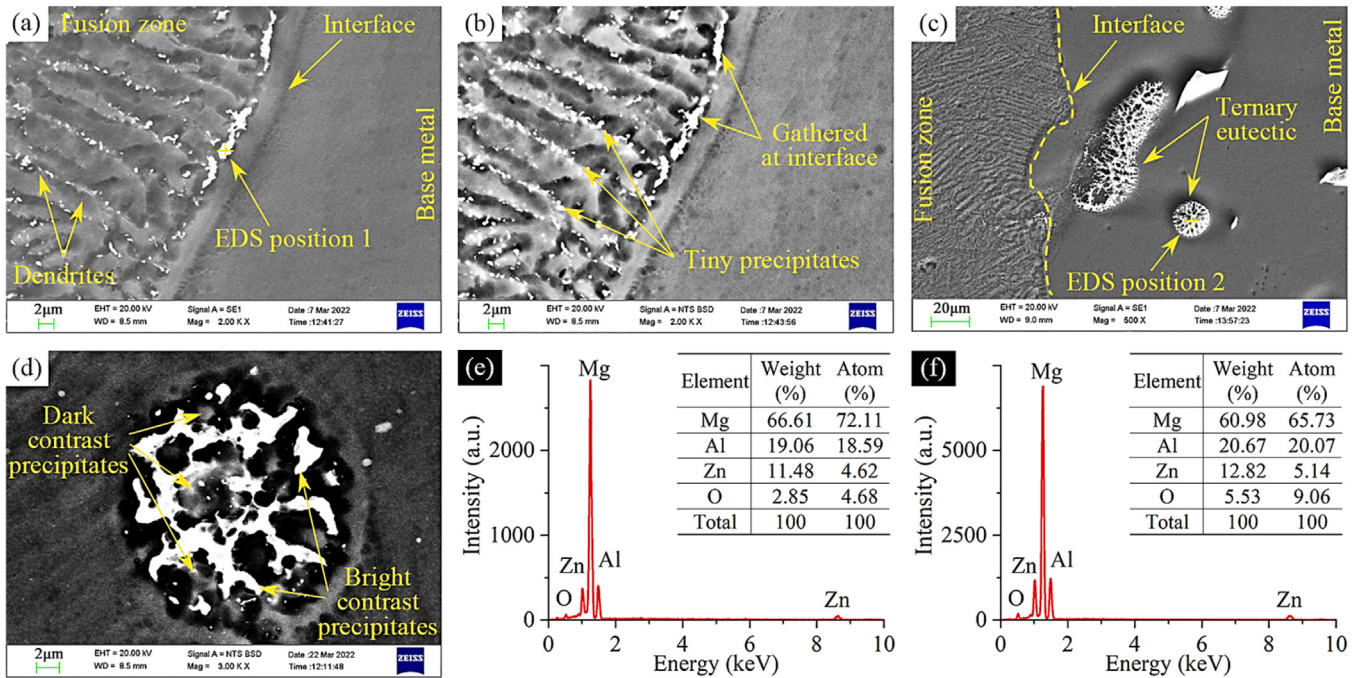


Fig. 12. The SEM and EDS results of the HAZ. (a) Precipitates near the bonded interface, (b) BSE image of the precipitates, (c) precipitates in the HAZ, (d) BSE image of the coral-like precipitates, (e) EDS spectrum of position 1, (f) EDS spectrum of position 2 [19].

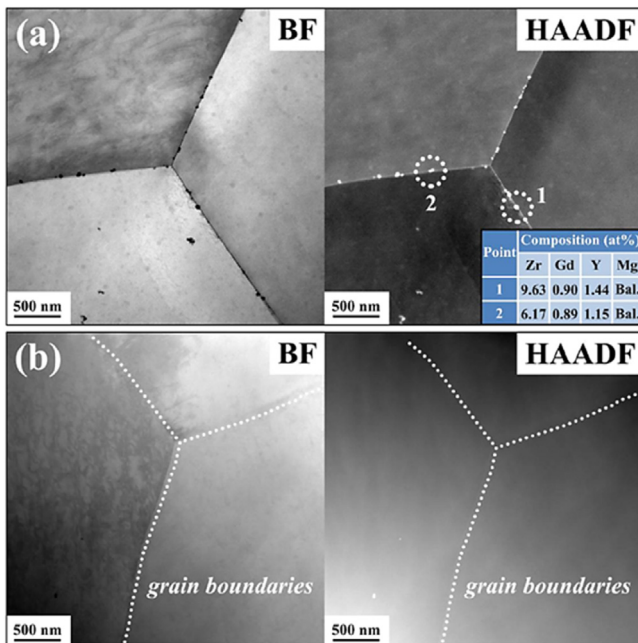


Fig. 13. TEM images of the secondary particles pinning condition at the GBs of the solution treated (a) VW63K FZ and (b) VW63 FZ [110].

by elevating dislocation density. Nevertheless, attention must be paid to low-melting-point conventional compounds (e.g., $\text{Mg}_{17}\text{Al}_{12}$), whose eutectic liquid phases formed under thermal cycles tend to accumulate at GBs. These phases exhibit elastic modulus mismatch with the matrix, forming brittle second phases that may act as potential crack initiation sites [20,60, 95,130]. It is noteworthy that the grain refinement and

second-phase strengthening induced by rare earth elements remain significant even under the rapid solidification conditions triggered by high thermal diffusivity. This demonstrates that the addition of RE elements provides an effective metallurgical pathway to overcome the welding challenges posed by the inherent physical properties of magnesium alloys. Future investigations should focus on optimizing RE element ratios to strike a balance between GBs pinning efficiency and second phases toughness, thereby providing theoretical foundations for engineering high-performance Mg alloy welded joints.

4. The generation mechanism of laser welding defects and RE regulation

As a high-energy beam processing technology, laser welding still faces challenges such as porosity and cracking in magnesium alloy joining. It is noteworthy that the inherent low specific heat capacity and high thermal diffusivity of magnesium alloys intensify the characteristic "rapid heating and rapid cooling" process of laser welding. This not only heightens the degree of non-equilibrium solidification but also induces high residual stresses due to substantial thermal gradients, thereby significantly increasing susceptibility to defect formation. This chapter will examine the formation mechanisms of porosity and cracks separately, with particular emphasis on the regulatory role of RE elements.

4.1. Porosity defects

4.1.1. The influence of heat input on porosity

The keyhole, as the most critical dynamic structure in laser welding processes, predominantly governs weld formation and

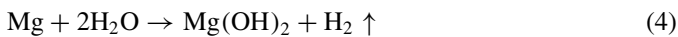
porosity generation [43,106, 131]. The formation mechanisms of process-induced pores primarily encompass four types: hydrogen precipitation, gas entrapment caused by turbulent liquid flow on the molten pool surface, keyhole collapse, and evaporation of high-vapor-pressure elements with low boiling points [132–134].

Of particular significance, the formation of hydrogen-induced porosity is intrinsically linked to hydrogen solubility in Mg [135]. Crucially, the dynamic behavior of the keyhole—a defining feature of laser deep-penetration welding—exhibits strong dependence on heat input variations. Excessive heat input elevates molten pool temperature, which triggers vigorous vaporization of low-boiling-point elements. This process generates localized high-pressure vapor bubbles that disrupt molten pool equilibrium, consequently inducing spatter and porosity formation. Therefore, the four process-induced porosity mechanisms can be categorized into two primary classes: hydrogen precipitation, and welding heat input coupled with keyhole stability.

(1) Hydrogen precipitation

In laser welding of Al alloys, hydrogen segregation plays a negligible role in large pore formation within weldments [136]. However, contrasting with Al alloys, molten Mg exhibits substantially higher hydrogen solubility compared to solid Mg state. The solubility of hydrogen in molten Mg increases significantly with rising temperature, reaching approximately 30 ml/100 g at 730 °C. When the molten pool solidifies, this solubility plummets to less than one-tenth of this value [137,138]. This supersaturated condition drives hydrogen to precipitate as molecular H₂, forming gas pores. Consequently, hydrogen-induced porosity from H evolution constitutes a critical challenge in Mg alloy welding processes [139].

Hydrogen sources during welding are multifarious. Residual hydrogen inherited from BM or filler wire during prior melting processes becomes liberated within high-temperature molten pools. Additionally, both water molecules adsorbed on surface oxide layers and those captured through molten pool-atmospheric interactions undergo chemical reactions to generate H₂, as shown in Eq. (4). Research work by J.D. Shearouse et al. [140] revealed that during the final solidification stage of AZ91 Mg alloy, H₂ liberation from the Mg₁₇Al₁₂ intermetallic phase directly facilitates the nucleation and propagation of micropores.



Although optimizing welding parameters can partially facilitate hydrogen escape from the molten pool, inherent limitations persist due to the strong adsorption of moisture on the highly reactive Mg alloy surface and the rapid solidification characteristics of the molten pool. Consequently, process parameter adjustments exhibit intrinsic constraints in suppressing hydrogen porosity, necessitating breakthroughs in alloy compositional design.

(2) Welding heat input coupled with keyhole stability

The keyhole within the molten pool constitutes the defining characteristic of laser deep penetration welding. As the piv-

otal process parameter, welding heat input directly governs keyhole dynamics, including depth, oscillation, and collapse propensity. These behavioral aspects, in turn, dictate intense fluid flow, thermal distribution, and gas entrainment within the molten pool, ultimately exerting a profound influence on overall pool stability and pore formation [141].

Fundamentally, keyhole stability is dictated by the dynamic equilibrium among recoil pressure, molten pool surface tension ($\gamma \cdot \kappa$, where γ is the liquid surface tension and κ is the principal curvature), and molten pool hydrostatic pressure ($P = \rho gh$, where ρ is the liquid density, g is gravitational acceleration, and h is the keyhole depth). Recoil pressure, generated by the high-velocity ejection of laser-induced metal vapor, serves as the primary driving force sustaining keyhole openness. Therefore, the presence of low-boiling-point elements (e.g., Zn, Al) in the material markedly elevates the risk of keyhole instability [25,43, 46]. Strong recoil pressure and vapor expansion effects severely perturb and can even disrupt the mechanical equilibrium at the keyhole opening and neck regions [134]. Direct consequences include pronounced keyhole oscillation, molten droplet spatter, and the induction of complex molten pool flows: This establishes conditions favorable for gas entrainment and subsequent entrapment [107,142–145].

Molten pool flow field characteristics are pivotal determinants of bubble trajectory and ultimate escape potential. To gain deeper insight into bubble motion behavior within a perturbed flow field and its association with pore formation, Yang et al. [146] comparatively investigated molten pool flow fields and bubble forces under two distinct welding modes, single laser welding (SLW) and ARM laser welding, as illustrated in Fig. 14. Within the molten pool, bubbles are primarily subjected to buoyancy force (F_b), viscous resistance (F_v), and fluid dynamic pressure (F_d), the direction of which varies with the flow field, as defined by Eqs. (5), (6), and (7):

$$F_d = \frac{1}{2} \rho_l \pi R^2 C_D (u_l - u_b)^2 \quad (5)$$

$$F_b = \frac{4}{3} \pi R^3 (\rho_l - \rho_b) g \quad (6)$$

$$F_v = \frac{9}{8} \rho_l \pi R^2 (u_l - u_b)^2 \quad (7)$$

Where, R is the bubble radius, C_D is the drag coefficient; u_l is the velocity of the molten pool around the bubble, u_b is the velocity of the bubble, ρ_l is the density of the fluid; ρ_b is the density of the bubble.

Model results indicate substantially higher molten pool flow velocities in SLW (0.25~0.75 m/s) compared to ARM (0~0.5 m/s). In high-velocity SLW molten pools, bubbles experience increased magnitudes of F_b and F_v , and are predominantly propelled backward and upward by hydrodynamic forces. Fig. 14(c) further reveals frequent directional shifts in SLW molten pool vortices. These shifting vortex directions drive bubbles either backward-downward or forward-downward, heightening their risk of capture by the solidi-

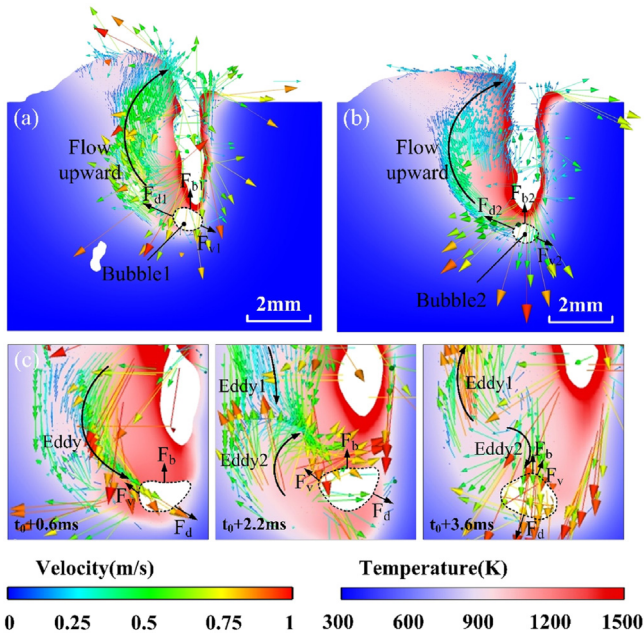


Fig. 14. Schematic diagram of force analysis of bubbles: (a) and (b) are the force state of bubbles in the molten pool the steady state of the SLW and ARM welding processes, respectively; (c) the force state of the bubbles when the flow direction of the molten pool changes during SLW welding [146].

fication front and pore formation. In contrast, within low-velocity ARM molten pools, diminished F_b and F_v magnitudes result in slower bubble movement. Stable vortex orientation favors bubble ascent and escape. Furthermore, annular beam integration significantly enlarges the keyhole diameter along its depth, thereby providing bubbles with larger pathways and shorter return routes to either the keyhole cavity or molten pool free surface, ultimately enhancing successful escape probability.

Consequently, the compatibility between keyhole morphology/stability and molten pool flow field patterns is critical in determining successful bubble escape. RE significantly enhance flow field stability by reducing both the liquidus and solidus temperatures and melt viscosity of Mg-RE alloys [32,147]. This establishes the physical basis for RE suppression of process-induced porosity. Advanced welding techniques, such as pulsed laser welding and ARM laser welding, derive their core advantage from precisely controlling welding heat input characteristics. This enables optimization of keyhole geometry, stabilization of the molten pool flow field, and extension of the bubble escape window period. Research by Xin et al. [106] on pulsed laser-GTA hybrid welding confirmed that excessively high peak pulse power amplifies keyhole oscillation and elevates pore content, whereas extending the keyhole opening duration provides a longer temporal window for bubble escape, reducing gas entrapment. Additionally, appropriately increasing welding speed lowers heat input per unit area, mitigating molten pool overheating and beneficial for sustaining keyhole stability. Quantitative results from Lu et al.'s [148] 3D coupled model further support this viewpoint: increasing laser power, reducing travel speed, or decreasing beam diameter elevates local heat input density

or molten pool lifetime, markedly raising keyhole collapse frequency and consequently increasing pore content. Consequently, developing novel welding techniques while optimizing welding heat input parameters constitutes the fundamental approach to achieving keyhole stability, improving molten pool flow fields, facilitating bubble egress, and ultimately mitigating weld pore formation.

4.1.2. Primary porosity

Primary porosity originates from pre-existing gas entrapment defects within the BM of die-cast Mg alloys. During the die-casting process, high-pressure rapid mold filling prevents complete evacuation of gases (e.g., N_2) from the mold cavity, resulting in micron-scale high-pressure gas clusters embedded in the matrix [149,150]. During welding, intense turbulent flow in the molten pool promotes gas coalescence, thereby generating macroscopic porosity. These defects generate severe joint flaws that significantly deteriorate joint quality [151].

In the study by M. Pastor et al. [152] on continuous-wave laser welding of die-cast AM60B, macroporosity in the weld pool primarily originated from the expansion and coalescence of pre-existing pores in the BM. Optimizing welding parameters by increasing welding speed and reducing laser power decreased macroporosity to levels comparable with BM porosity, whereas enlarging beam defocusing distance showed no discernible pore reduction. Similarly, Zhao et al. [153] investigated continuous-wave Nd:YAG laser welding of die-cast AM60B, demonstrating that re-melting welding substantially reduces porosity. During the initial welding pass, rapid solidification restricted bubble escape, while secondary re-melting provided extended opportunities for bubble floatation. Beyond temporal factors, these bubbles may coalesce into larger entities under molten metal flow during re-melting. According to Archimedes' principle, larger bubbles experience greater buoyant forces and ascend more rapidly, significantly enhancing evacuation efficiency. Wang et al. [154] conducted TIG welding research on die-cast AZ91D, revealing that incorporating extruded AZ61 filler wire markedly reduced pore quantity and diminished macroporosity dimensions. This is attributed to the fact that the added extruded AZ61 alloy itself has almost no pore defects, which "diluted" the pores in the molten pool after addition.

In laser welding of die-cast Mg alloys, porosity defects in weld joints primarily originate from the porosity inherent in the BM. The influence of keyhole stability on pore formation is relatively limited [153]. This finding demonstrates that for die-cast Mg alloys, controlling intrinsic porosity should take precedence over optimizing processing parameters. Therefore, a multi-stage control strategy is recommended to address such defects.

The foremost and most effective approach lies in optimizing die-casting processes to minimize gas-entrapped porosity within the BM. Subsequent measures include appropriately increasing heat input or applying remelting techniques to prolong the molten pool's high-temperature duration, thereby reducing solidification rates and providing enhanced kinetic

conditions for bubble escape [134]. Ultimately, the introduction of extruded welding wires with low porosity concentrations can diminish overall porosity levels in the molten pool through dilution effects, while simultaneously regulating molten pool flow patterns to suppress macroscopic porosity aggregation.

4.2. Crack defects

4.2.1. The influence of heat input on cracks

As a representative high-temperature failure mode, welding hot cracks predominantly initiate within the brittle temperature range (BTR) near the material's solidus temperature, exhibiting characteristic intergranular propagation along GBs [155]. Laser power and welding velocity emerge as predominant factors governing cracks formation and propagation in laser welding processes. Research on AZ91D laser welding by Zhou et al. [156] revealed an increased crack propagation length within weld metal under elevated laser power conditions. Notably, when welding velocity is enhanced, crack dimensions can be reduced under both high and low laser power regimes. At lower power levels, appropriately reducing the welding speed optimizes heat input to balance the filling effect of low-melting-point eutectic liquid films and stress relaxation mechanisms. This approach simultaneously mitigates cooling rates, diminishes thermal stresses, and potentially achieves complete crack suppression.

Beyond this simple correlative relationship, the heat input—directly governed by laser power and welding speed—dictates molten pool penetration depth, thereby influencing susceptibility to solidification cracking [157–160]. In TIG welding research conducted by Chen et al. [157], increasing the welding speed to 6 mm/s resulted in incomplete penetration. Despite this, crack susceptibility unexpectedly decreased. Conversely, elevating the current to 160 A achieved full penetration, accompanied by an abrupt reduction in crack incidence. This phenomenon arises from distinct mechanical constraints under varying penetration states. At lower heat input, incomplete penetration retains substrate constraints at the workpiece bottom. However, the diminished molten pool dimensions and moderated thermal gradients alleviate stress concentration in shallow regions. As heat input increases, the welding mechanism transitions from conduction-dominated to keyhole mode, progressively increasing penetration depth. Full penetration eliminates substrate constraints during solidification, enabling global stress relaxation to dominate over localized stress accumulation, thereby suppressing crack formation. These findings demonstrate that crack susceptibility under incomplete penetration exceeds that in fully penetrated conditions. The synergistic interplay between low strain rates and restricted molten pool geometry effectively inhibits both crack initiation and propagation.

While reducing heat input proves effective for crack suppression, the non-linear characteristics between welding cracks and heat input arise from its dual dependence on laser power and welding speed. In Zhou et al.'s research [156], solidification cracks propagated along GBs when using 250 W

laser power with 10 mm/s welding speed. Remarkably, identical heat input conditions (25 J/mm) with reduced parameters (50 W laser power at 2 mm/s) achieved complete crack elimination.

More critically, the inherently high thermal diffusivity of magnesium alloys is a key factor that must be considered behind this nonlinear relationship. The high thermal diffusivity leads to extremely rapid cooling in the weld seam and HAZ. This not only intensifies the non-equilibrium characteristics of the solidification structure but also generates significant tensile stresses due to hindered contraction, thereby aggravating the formation of hot cracks.

To address the compatibility issue among welding heat input, process parameters, and the high thermal diffusivity of magnesium alloys, Zhao et al. [134] implemented preheating treatment on workpieces to mitigate welding-induced thermal stresses. Research results demonstrated that preheating elevates molten pool temperatures, moderates cooling rates, and suppresses non-equilibrium crystallization and intergranular liquid film formation, effectively reducing crack susceptibility. Complete crack elimination was achieved at a 200 °C preheating temperature. Concurrently, preheating optimizes molten pool recoil pressure dynamics, diminishing turbulent flow in liquid metal and ameliorating undercut defects. TIG welding parameters govern molten pool cooling and stress by controlling heat input. The incorporation of a filler wire utilizes endothermic melting to lower net heat to the workpiece and improve pool fluidity, thereby mitigating crack formation risks [157–161]. This mechanism was confirmed in Yu et al.'s study [160] on ZK60 magnesium alloy using a ZK40 filler wire. The approach reduced effective heat input, moderated thermal gradients, and counteracted rapid cooling from the alloy's high thermal diffusivity, significantly diminishing crack susceptibility and producing crack-free welds.

4.2.2. Effect of Al and Zn contents on hot cracking susceptibility

As schematically illustrated in Fig. 15, the PMZ corresponds to the semi-solid (S + L) region adjacent to the FZ. Within the HAZ, regions experiencing peak temperatures between the alloy's liquidus and effective solidus temperatures undergo constituent segregation and liquefaction at GBs, forming interfacial liquid films. This significantly reduces the cohesion and strength of the S + L region.

Subsequently, during solidification and cooling, tensile stresses develop in the PMZ from two primary sources: (1) solidification shrinkage associated with the liquid-to-solid (L→S) transition of the weld metal, and (2) differential thermal contraction induced by coefficient of thermal expansion (CTE) mismatch. When these combined tensile stresses exceed the weakened intergranular strength of the S + L structure, the liquid films rupture, initiating and propagating cracks along the embrittled GBs.

In summary, liquation cracking involves two critical stages: liquid film formation during heating and liquid film rupture during solidification/cooling. Further, this analysis reveals that Mg alloy's hot cracking susceptibility is governed not only

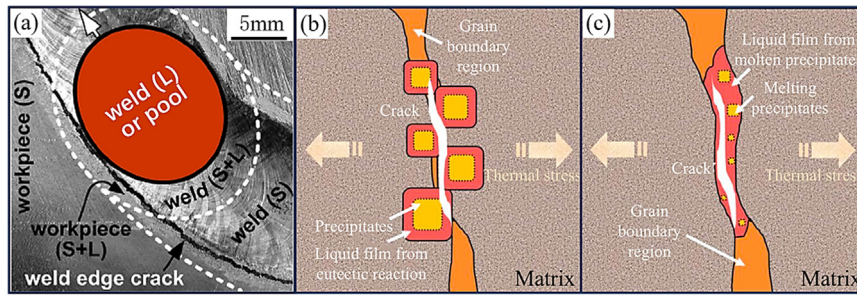


Fig. 15. (a) A schematic weld pool superimposed on a weld to show liquation cracking along the outer weld edge, weld (S + L), and workpiece (S + L). The white arrowhead indicates the travel direction of the weld pool during welding [42]. Schematic of the liquation cracking caused by (b) constitutional liquation and (c) direct phase melting [162].

by welding parameters but also by physical properties including CTE and thermal conductivity, coupled with the content and species of low-melting-point second phases [156,158,163,164]. Originating from the low melting temperatures of intermetallic compounds like $\text{Mg}_{12}\text{Al}_{17}$ (470 °C) and MgZn_2 (340 °C), the cracking propensity exhibits a positive correlation with Al and Zn concentrations.

Investigation by Huang et al. [158] demonstrates that under single-pass welding thermal cycles, cracking in AZ31 alloy predominantly localizes within the FZ, with limited propagation into the HAZ. In contrast, high-Al AZ61 alloy exhibits preferential crack distribution within the weld metal HAZ due to grain boundary segregation of low-melting $\beta\text{-Mg}_{17}\text{Al}_{12}$ phase, demonstrating a total crack length (TCL) of 10.50 mm, significantly higher than that of AZ31. Remarkably, as the number of thermal cycles increased to three times, the TCL of AZ61 decreased by 1.27 mm. This reduction is attributed to Al content depletion in grain boundary liquid alloys from 13.3 wt% to 7.7 wt% during multiple high-temperature exposures, which suppresses persistent liquefaction of $\text{Mg}_{17}\text{Al}_{12}$ phase while narrowing the effectively liquefied width in the PMZ.

Analogous phenomena have been corroborated by the research of Chen et al. [164] and Zhou et al. [156]. Notably, Shen et al. [163] demonstrated that in the double-sided TIG welding of AZ61, the secondary thermal cycle led to the remelting of the β phase formed for the first time, resulting in a sudden drop in grain boundary strength. Even a residual stress of only 45 MPa could trigger intergranular liquefaction cracks.

The apparent paradox regarding the influence of multiple thermal cycles on crack susceptibility in magnesium alloy welds underscores the dual nature of its underlying mechanism, profoundly highlighting the high dependence of weld quality on process parameters. In the study by Huang et al. [158], the reduced crack susceptibility is attributed to the decreased Al content within the molten pool during repeated high-temperature exposure, which consequently weakens the effective liquation width in the PMZ. Conversely, Shen et al. [163] demonstrated that excessively high peak temperatures in the thermal cycle, surpassing the melting point of the $\beta\text{-Mg}_{17}\text{Al}_{12}$ phase, lead to its remelting. This results in the formation of continuous liquid films along GBs, significantly

degrading GBs strength and thereby inducing cracking. The welding process parameters intrinsically govern the thermal cycle's temperature profile, particularly the peak temperature. Therefore, rationally designing multi-pass welding sequences or heat treatment processes to confine thermal cycles within an appropriate temperature range can exploit their beneficial regulatory effects. In contrast, improper thermal cycle parameters may intensify the cracking tendency and jeopardize weld integrity.

In autogenous welding techniques analogous to laser welding, the reduction of Al and Zn contents relies solely on the evaporation of elements at high temperatures. To regulate Al/Zn levels, different alloys can be employed as filler materials to mitigate crack susceptibility. In the laser welding research on ZK60 conducted by Yu et al. [160], the incorporation of ZK40 filler strips reduced Zn content in the weld region while absorbing a portion of the thermal energy, thereby decreasing the effective heat input to the workpiece and ultimately achieving crack-free welds. In research by Yuan et al. [42], the Al content in filler wires, workpieces, and patch materials influences weld toe crack susceptibility by affecting compositional differences and solid fraction (f_s) disparities. As illustrated in Fig. 16, leveraging the phase diagram and weld microstructure analysis of alloy Co, the authors established a predictive criterion for fusion boundary crack formation: the Al content differential constitutes the dominant variable governing weld crack susceptibility, and a critical threshold is established wherein pronounced weld cracking manifests when the solid fraction disparity between the weld zone and base material exceeds 0.05 ($f_s^{\text{weld}} - f_s^{\text{workpiece}} > 0.05$). In the corresponding research, Medium-high Al filler wires significantly reduce cracking risks when matching the Al content of the workpiece/patch, while low-Al fillers increase cracking risks, especially with high-Al substrates. Although low-Al materials have narrower solidification ranges and less liquid phase, cracks may still form if the weld's Al content exceeds the workpiece's ($f_s^{\text{weld}} > f_s^{\text{workpiece}}$).

Consequently, solidification cracks in molten pools readily act as crack nucleation sites, propagating along liquefied GBs in the HAZ to form through-thickness cracks. The formation of HAZ liquation cracks is intrinsically linked to stress concentration at weld toe regions and localized thermal cycling. Both crack types fundamentally originate from wide

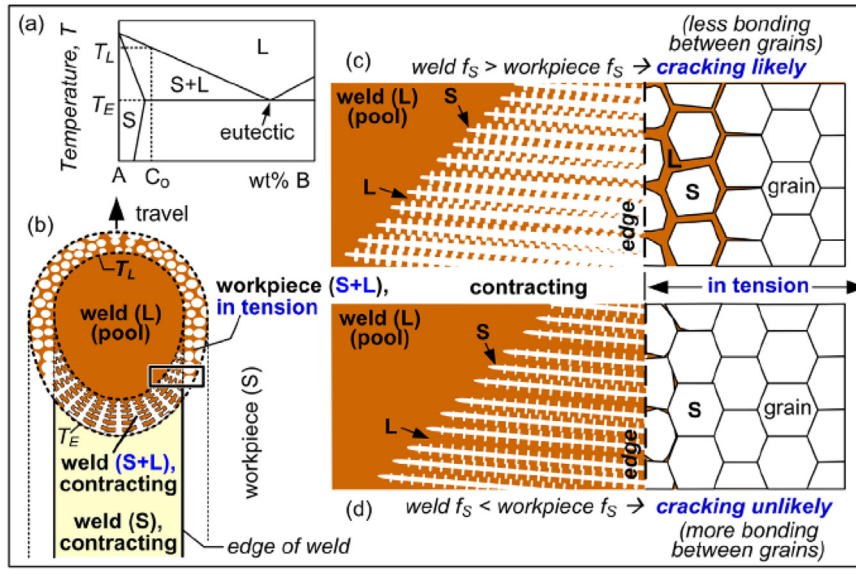


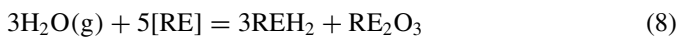
Fig. 16. Susceptibility to cracking along the weld edge: (a) phase diagram showing alloy C_0 (workpiece); (b) microstructure around weld pool; (c) cracking likely; (d) cracking unlikely. Rectangular box in (b) is enlarged in (c) and (d) [42].

solidification temperature ranges and low-melting eutectic liquid films. Therefore, effective mitigation requires balancing heat input with cycling frequency to simultaneously suppress excessive Al vaporization and ensure sufficient liquid phase infiltration along GBs, thereby preventing cumulative microcrack networks from cyclic remelting. Post-weld heat treatment or mechanical vibration can eliminate residual stresses while reducing grain boundary separation driving forces, synergistically enhancing crack resistance.

4.3. Suppression of defects by RE elements

4.3.1. The influence of RE elements on porosity

As critical alloying constituents in Mg alloys, RE elements offer a novel approach to hydrogen porosity control through their unique physicochemical interactions with hydrogen. RE elements demonstrate exceptional hydrogen dissolution capabilities, enabling hydrogen atom capture via surface adsorption even at room temperature, with hydrogen absorption capacity intensifying markedly at elevated temperatures. For instance, at 800 °C, cerium (Ce) can dissolve hydrogen up to 14,500 mL/100 g, while lanthanum (La) achieves 14,300 mL/100 g, far exceeding the hydrogen accommodation capacity of the Mg matrix [165,166]. When RE-containing Mg alloy molten pools interact with environmental H_2O or H_2 , preferential chemical reactions occur between RE elements and these hydrogen sources [167,168], as illustrated in Eqs. (8) and (9).



RE elements effectively reduce the free hydrogen content in liquid metal through chemisorption and solid solution

mechanisms. Huang et al. [169] investigated hydrogen adsorption at GBs by different solute atoms through first-principles calculations, with results presented in Fig. 17. Computational data reveal that lower formation energies of adsorption configurations (SH_x) indicate stronger hydrogen adsorption capability of the solute atom. Fig. 17 demonstrates that RE elements generally exhibit superior hydrogen adsorption capacity. Compared with the Mg matrix, hydrogen atoms preferentially accumulate around RE solute atoms with high adsorption capability, forming stable hydride clusters at twin boundaries (TBs). This confirms that solute atom segregation at grain boundaries exhibits heightened hydrogen adsorption, establishing these regions as preferential sites for SH_x formation. Consequently, introducing rare earth elements into the alloy creates high-density hydrogen traps at GBs and phase interfaces. These traps significantly impede long-range hydrogen migration while obstructing hydrogen bubble nucleation and growth [170–172].

Moreover, unexpectedly, Zr exhibits stronger adsorption capability than Nd and Y. In the research on laser welding using sand-cast ZE41A-T5 alloy by Cao et al. [43], Zr reacted with hydrogen to form ZrH_2 , which reduced hydrogen porosity and thereby significantly suppressed hydrogen pore formation. However, it is critical to note that ZrH_2 undergoes near-complete decomposition at 500~750 °C, releasing trapped hydrogen atoms. Consequently, weld joints relying on Zr for hydrogen immobilization cannot undergo post-weld high-temperature heat treatment to avoid hydrogen reprecipitation [173,174].

Beyond hydrogen porosity suppression, RE elements also facilitate the escape of process-induced pores generated under high-energy-density welding conditions. In Mg alloys, an appropriate amount of RE additions lowers both the liquidus and solidus temperatures of Mg-RE alloys [32,147]. This phe-

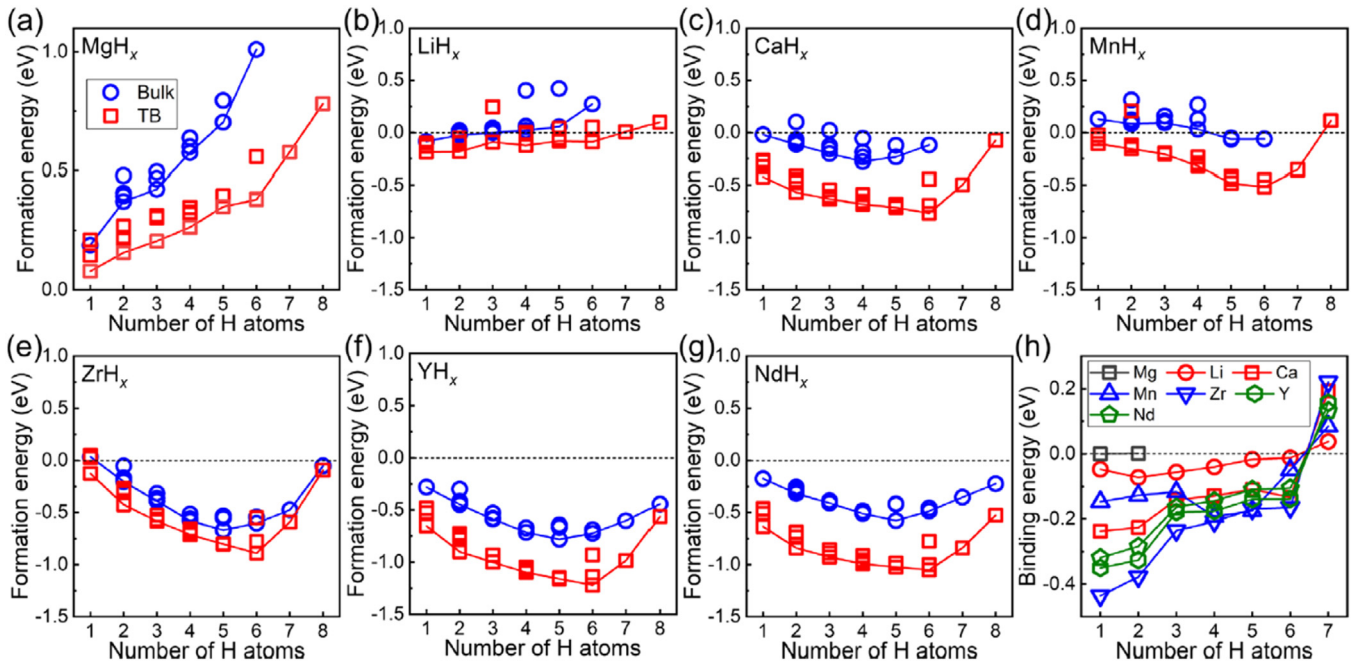


Fig. 17. Formation energies of (a) MgH_x and (b-h) SH_x clusters in bulk Mg and twin boundary. (h) Variation of binding energy with the number of H atoms in a MgH_x or SH_x cluster located in a TB [169].

nomenon can enhance the fluidity of the molten pool under equivalent thermal conditions, which increases the probability of bubbles escaping from the viscous melt, thereby reducing the overall porosity density.

To sum up, the addition of RE and Zr to Mg alloys serves as a critical method for suppressing porosity defects. Beyond alloy composition design, differentiated process optimization strategies must be implemented according to specific laser welding modes. In conduction mode welding, where unstable keyhole structures have not formed in the molten pool, priority should be given to incorporating elements such as Zr, Nd, Ce, and La. These elements achieve hydrogen immobilization, inhibiting hydrogen evolution while forming high-melting-point precipitates, thereby reducing the evaporation of low-melting-point elements. Conversely, in keyhole mode welding, where porosity primarily originates from keyhole instability, a dual approach is required. Beyond hydrogen control and suppression of low-boiling-point element evaporation, it is essential to prolong keyhole opening duration or enlarge keyhole diameter through optimization of welding heat input and approaches that refine the spatiotemporal distribution of laser energy, including pulsed lasers or adjustable annular beams. These measures collectively enhance keyhole stability while balancing molten pool dynamics and the evaporation effects of elements.

4.3.2. The influence of RE elements on cracks

Whether it is solidification cracking or liquefaction cracking, the fundamental cause lies in the inability of residual metallic liquid films—formed by low-melting-point eutectic phases (e.g., Mg-Al, Mg-Zn)—to withstand localized stresses induced by solidification shrinkage during cooling. Conse-

quently, optimizing alloy composition to reduce the content of these low-melting-point eutectic phases constitutes the primary approach for suppressing welding cracks and mitigating the alloy's cracking susceptibility. Simultaneously, refining welding parameters to regulate heat input offers an auxiliary pathway to ameliorate crack-related defects. By synergistically controlling compositional design and process conditions, the mechanical integrity of welded Mg alloys can be significantly enhanced through dual suppression of metallurgical and thermomechanical cracking mechanisms.

Li et al. [85,175] demonstrated that the addition of Ce to AZ-series Mg alloys resulted in the formation of a new Al_4Ce phase, which reduced the content of $\beta\text{-Mg}_{17}\text{Al}_{12}$ phase and refined weld grain structure. Building upon this work, Cai et al. [176] introduced Ce, Y, and Gd RE elements into AZ91 alloy. The RE additions significantly decreased the content and refined the size of the $\text{Mg}_{17}\text{Al}_{12}$ phase, with Ce exhibiting the most pronounced effect—reducing the $\text{Mg}_{17}\text{Al}_{12}$ phase content by 30 %. The RE elements reacted with Al to form high-melting-point compounds such as Al_4Ce , Al_2Y , and Al_2Gd , which suppressed the formation and aggregation of brittle $\beta\text{-Mg}_{17}\text{Al}_{12}$ phases and inhibited their coarsening. Consequently, crack initiation sites transitioned from coarse, continuous network structures to dispersed particulate phases, effectively hindering crack propagation.

In Al alloy welding, the property of forming Al-RE compounds with RE elements and Al elements is also often utilized for performance regulation. When preparing 7075 alloy using laser powder bed melting technology, the addition of Er can completely eliminate cracks [177]. This is attributed to the formation of Al_3Er phases, which reduce stress concentration and liquid film segregation at GBs while promoting

more uniform thermal stress distribution, thereby suppressing intergranular crack propagation. Similarly, when mixed RE elements (Ce and La) are introduced into A357 Al alloy, the RE additions transform acicular silicon phases into granular morphology, mitigating stress concentration. Consequently, the fracture path shifts from transgranular (through silicon phases) to intergranular (along α -Mg boundaries), resulting in 25.9 % and 34.2 % enhancements in elongation for as-cast and T6-tempered conditions, respectively [178].

Additionally, the high-melting-point compounds formed by RE elements exhibit resistance to cracking risks induced by welding thermal cycles. In the researches conducted by Chen et al. [19,20], phases such as $\text{Al}_{17}\text{Ce}_2\text{Mn}_7$ and $\text{Al}_{11}\text{Ce}_3$ remained thermally stable due to their high melting points, unaffected by welding heat input. In contrast, $\text{Mg}_{17}\text{Al}_{12}$ and MgZn_2 phases partially melted under elevated temperatures. However, due to their limited content, the resultant solidification shrinkage stresses were insufficient to initiate cracks.

Therefore, the addition of trace RE elements serves as an effective strategy to reduce the volume fraction of low-melting-point eutectic phases, suppress grain boundary liquation, and mitigate welding cracks. The formation of high-melting-point eutectic phases (e.g., Al_3Er , Al_4Ce) by RE elements refines the grain structure and reduces secondary dendrite arm spacing (SDAS), thereby minimizing crack initiation sites at GBs [176–179]. Simultaneously, these phases optimize phase morphology by suppressing the formation of continuous brittle networks and promoting granular or fibrous phase distributions, which effectively mitigates stress concentration [164,180, 181]. Additionally, Mg alloys' high thermal conductivity (156 W/(m·K)) and low specific heat capacity (1025 J/(kg·K)) lead to rapid post-weld cooling, generating significant residual thermal stresses. The incorporation of RE elements indirectly mitigates solidification shrinkage stresses in the liquid phase, while simultaneously obstructing crack propagation pathways through enhanced phase stability and strain accommodation.

In addition, while rapid solidification during welding had negligible impact on the overall β - $\text{Mg}_{17}\text{Al}_{12}$ content, it refined the β -phase size and disrupted its continuous distribution, thereby reducing stress concentration at GBs and minimizing crack nucleation sites [176]. The synergistic interaction between rapid solidification and RE additions further enhanced grain refinement, collectively lowering the alloy's susceptibility to welding cracks.

Thermodynamic analysis reveals that the key mechanism by which RE additions reduce low-melting-point eutectic phase content in Mg alloys lies in the preferential reactivity of RE elements with Al [96,182]. High-melting-point RE compounds preferentially precipitate only when the Gibbs free energy ($\Delta G_{\text{RE-Al}}$) for RE-Al reactions is lower than that of Mg-Al reactions (e.g., $\text{Mg}_{17}\text{Al}_{12}$ eutectic phase formation, $\Delta G_{\text{Mg-Al}}$). This thermodynamic hierarchy effectively suppresses crack susceptibility during alloy solidification [183,184]. The reaction free energy (ΔG) can be quantified using classical thermodynamic formulations, as shown

in Eq. (10) [182]:

$$\Delta G = \Delta H - T \Delta S \quad (10)$$

Where ΔH represents the enthalpy change of the reaction, ΔS denotes the entropy change of the reaction, and T is the thermodynamic temperature.

Despite these advantages, excessive RE content may excessively deplete solutes such as Zn and Al, reducing the eutectic liquid film during late-stage solidification. This diminishes interdendritic feeding capacity and increases hot tearing susceptibility [185]. Furthermore, surplus RE elements can promote the aggregation of RE-rich phases at GBs, forming coarse second phases that act as new crack initiation sites [178,185].

The optimization of process parameters and alloy composition exhibits synergistic effects. The former mitigates segregation and liquation risks by controlling peak molten pool temperature, cooling rate, and stress state. The latter fundamentally reduces low-melting-point phase content and strengthens GBs through RE element-regulated the microstructural. These dual strategies synergistically suppress the initiation and propagation of both solidification cracks and liquation cracks, ultimately enhancing the cracking resistance of Mg alloy welded joints.

5. Challenges and prospects

5.1. Challenges

The pivotal role of RE elements in laser welding of Mg alloys is primarily manifested through two interrelated mechanisms: microstructural regulation and defect suppression. In terms of microstructural control, RE elements refine FZ grains via the formation of high-melting-point intermetallic compounds and heterogeneous nucleation effects, while simultaneously pinning HAZ GBs to inhibit the growth of coarse columnar crystals and mitigate grain boundary coarsening. This dual action enhances joint microhardness and tensile strength, thereby significantly improving the mechanical performance of welded joints. Regarding defect suppression, RE elements effectively suppress hydrogen porosity by chemically adsorbing and immobilizing hydrogen atoms, reduce molten pool surface tension to facilitate gas escape, and strategically redistribute low-melting-point eutectic phases to minimize thermal crack propagation. These combined effects substantially enhance weld integrity by addressing both porosity and crack-related defects.

However, the extreme cooling rates inherent to laser welding introduce critical challenges. In the FZ, rapid solidification promotes interdendritic segregation of RE elements, creating localized enrichment zones that impede the full precipitation of high-melting-point compounds. Concurrently, within the HAZ, transient thermal cycles disrupt the equilibrium between dissolution and reprecipitation of the second phases, leading to microstructural instability. Compounding these issues is the current lack of quantitative models describing the interplay between welding heat input and RE elements behavior, which forces process optimization to rely heavily on

empirical trial-and-error approaches. These limitations hinder precise control over microstructure evolution and defect suppression.

Therefore, while RE strengthening mechanisms have proven effective in enhancing the welding performance of Mg alloys, three critical challenges persist in current research:

- (1) The multiscale complexity of microstructural evolution: It manifested through nanoscale precipitation of RE compounds and grain boundary segregation in the FZ, alongside dynamic dissolution-precipitation of high-melting-point phases in the HAZ, which creates competing effects between grain refinement strengthening and grain boundary embrittlement. Meanwhile, the absence of a cross-scale correlation mechanism linking composition, heat input, and microstructural properties hinders systematic guidance for process optimization.
- (2) Insufficient adaptability to service environments poses critical risks: Under high-temperature ($>300\text{ }^{\circ}\text{C}$) or corrosive conditions (e.g., 3.5 % NaCl solution), localized RE phase aggregation triggers micro-galvanic corrosion and creep-induced cracking. However, the underlying failure mechanisms, including evolving electrochemical potential gradients at GBs and stress concentration at second phase-matrix interfaces, along with corresponding mitigation strategies, remain inadequately characterized.
- (3) Engineering application bottlenecks persist: Laser path planning for complex-shaped components and uniformity control of RE elements distribution poses significant technical challenges. Scalable production is further constrained by the high cost of ultra-pure RE welding wires and stringent precision equipment requirements. Compounding these issues, current quality control standards lack RE-specific criteria for evaluating strengthening effects, significantly impeding industrial adoption in automotive lightweighting and aerospace applications. These unresolved challenges collectively limit the full exploitation of RE-enhanced Mg alloys in advanced manufacturing sectors.

5.2. Prospects

To address these challenges, future research must adopt an integrated approach centered on "mechanistic understanding, process innovation, and engineering validation" to achieve synergistic advancements in both fundamental theory and applied technologies.

- (1) At the mechanistic level: Leveraging molecular dynamics simulations, and CALPHAD thermodynamic calculations will elucidate the segregation kinetics of RE atoms during molten pool solidification, their interactions with alloying elements like Al/Zn, and the threshold pinning effects on grain boundary migration. These insights will enable the development of a multi-physics coupled predictive model.

- (2) At the level of process innovation: At the process level, emerging techniques such as pulsed laser modulation and ultrasonic-assisted welding can facilitate the uniform precipitation of rare earth compounds and improve molten pool dynamics. In terms of alloy design, the ratio of rare earth elements in the Mg-RE-Zr system should be optimized to balance strength with crack resistance, thereby reducing reliance on high heat input. Furthermore, process development must leverage intrinsic material properties like the low specific heat capacity and high thermal diffusivity of magnesium alloys. Techniques such as laser-arc hybrid welding and preheating can be employed to better control the welding thermal field and mitigate issues associated with high cooling rates.
- (3) At the service performance enhancement and engineering validation level: Relying on machine learning, we need explore the collaborative process of a variety of welding technique and RE welding wires. Through multi-faceted performance characterization to clarify how RE precipitates regulate oxide film stability and alter crack propagation paths. A dual "RE strengthening + surface coating" protection strategy will be engineered to synergistically mitigate degradation, ultimately enabling their scalable implementation in advanced manufacturing sectors.

The introduction of RE elements has pioneered an innovative pathway for Mg alloy laser welding, transitioning from mere defect suppression to performance enhancement. However, unlocking their full potential demands profound understanding of three critical domains: non-equilibrium solidification behavior, complex service-induced failure mechanisms, and engineering-critical technologies. Through technological innovation, a comprehensive framework integrating fundamental theory, process optimization, and engineering application can be established. This holistic approach will provide critical support for designing and manufacturing high-performance lightweight welded structures, accelerating the widespread adoption of Mg alloys in aerospace, new energy vehicles, and other advanced sectors. Ultimately, such advancements will propel the industrial lightweighting initiative to more sophisticated stages, achieving synergistic gains in structural efficiency and sustainability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Lingxiao Wang: Writing – original draft. **Xiaoya Chen:** Writing – review & editing. **Quanan Li:** Writing – review & editing. **Zheng Wu:** Methodology. **Yuefei Dai:** Investigation. **Zeyu Zheng:** Investigation. **Kang Yao:** Investigation.

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